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Development of an optical scattering measurement device

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Produktutveckling av ett optiskt mätinstrument

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Sammanfattning

Optiska mätinstrument används för att mäta ljusreflektion och ljusspridning från material, för att erhålla data om materialets yt- och bulkegenskaper. Mätdata används ofta i forsknings- och utvecklingsprojekt där materialkrav är viktiga, även för kvalitetskontroll i tillverkningsprocesser, i olika optiska simuleringar och kan användas för fotorealistisk rendering. I detta examensarbete, genomfört på AFRY, kommer ett multifunktionellt team att utveckla en optisk spridningsmätningsenhet som syftar till att samla in data mer noggrant. Denna avhandling kommer att fokusera på den mekaniska designen av enheten som består av stabiliteten och rörelsen av komponenterna, mätmiljön och materialval med fokus på prestanda och hållbarhet.

Den optiska modellen som kommer att ligga till grund för utvecklingen kommer att vara en gonioreflektometer bestående av ett materialprov, provhållare, ljuskälla, detektor och en miljö där mätningarna genomförs. Några av de fysiska, kognitiva och emotionella behoven hos den avsedda användaren är effektiv användning, låg risk för felanvändning, pålitlighet och hög precision. En noggrann kravspecifikation gjordes som en ram för konceptgenereringen.

Det valda konceptet möjliggör rörelse av de optiska komponenterna med ett vinkelsteg som tillåter den önskade optiska spridningsmätningen. Den valda stegmotorn och utväxlingen ger flexibilitet i rörelsen, detta bidrar till att det är enkelt för användaren att ändra vinkelstegen för de optiska komponenterna, vilket tillåter både fina och grova mätningar. En avskiljningsskärm valdes för att undvika ljuskontaminering mellan mätningarna och materialprovhållaren liknar en ram där materialprovet monteras utanför enheten. Det mekaniska systemet har en hög stabilitet och materialet svart anodiserad aluminium bidrar till konstruktionens robusthet. En fysisk prototyp skapades för att validera rörelsen, eftersom rörelsen av detektorn och ljuskällan kommer att vara liknande, återskapades endast detektorns rörelse. Prototypen visade att detektorns rörelse fungerade på önskat sätt, därmed godkänns konstruktionen av rörelsen. Projektets mål avgränsades i samförstånd med projektmedlemmarna och handledarna på grund av tidsramen, därmed finns det framtida arbete för mätinstrumentet som bör beaktas. Sammanfattningsvis uppfylldes projektets syfte efter att målen avgränsats och en konceptuell lösning skapades som uppfyllde projektets krav.

Nyckelord: Optiskt mätinstrument, ljusspridningsmätning, BSDF, BRDF, BTDF, spekulärt och diffust ljus, ljusspridning, material ytjämnhet, ljusintensitet, optisk modell, ytegenskaper, materialegenskaper, mekaniskt system, sfärisk rörelse, sfärisk mätning, stegmotor, avskiljningsskärm, materialprov, transmission, reflektion, utväxling, prototyp, moodboard, glob, multifunktionellt projekt, användarvänlighet, tjänst

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Abstract

Optical scattering measurement devices are used to measure light reflection and light scattering from materials, to obtain data of the surface and bulk properties of materials. The measurement data are often used in research and development projects where material requirements are important, also for quality control in manufacturing processes, in different optical simulations and can be used for photorealistic rendering. In this master thesis project conducted at AFRY, a multifunctional team will develop an optical scattering measurement device that aims to collect data more accurately than current devices on the market. This thesis will focus on the mechanical design of the device which consists of the stability and movement of the components, the environment of the measurements and material selection with a focus on performance and sustainability.

The optical model that will act as a basis for the development will be a gonioreflectometer consisting of a material sample, sample holder, light source, detector and an environment in which the measurements are conducted. Some of the physical, cognitive and emotional needs of the intended user are efficient use, low risk of misuse, reliable and high precision. A thorough requirement specification was made as a framework for the concept generation.

The selected concept provides the movement of the optical components with an angular step enabling the desired optical scattering measurement. The selected stepper motor and gear ratio provides the flexibility of the movement, making it easy for the user to change angular steps of the optical components, enabling both fine and rough measurements. A separating screen was chosen for both concepts in order to avoid light contamination between measurements and the material sample holder resembles a frame that allows for mounting the material sample outside of the device. The mechanical system has a high stability and the material black anodized aluminum further contributes to the sturdiness of the construction.

A physical prototype was created to validate the movement, since the movement of the detector and light source will be similar, only the detector movement was prototyped. The prototype showed that the movement of the detector worked in the desired way, hence the construction of the movement is approved. The scope was delimited in consensus with the project members and supervisors due to the time frame, hence there is future work on the device that should be accounted for. In conclusion, the purpose of the project was fulfilled after delimiting the goals and a conceptual solution was created that fulfilled the requirements of the project.

Keywords: Optical scattering measurement device, light scattering measurement, BSDF, BRDF, BTDF, specular and diffuse light, light scattering, material surface roughness, light intensity, optical model, material surface properties, material bulk properties, mechanical system, spherical measurement, stepper motor, separating screen, material sample, gear ratio, prototype, moodboard, globe, multifunctional project, usability, service

FOREWORD

Now that the master thesis project at AFRY has come to an end, we want to send our gratitude to a number of people that have joined us in the process. The path to this point has been both challenging and rewarding and it would not have been possible without the collaboration, support and guidance by the following:

Thanks to our incredible project members Amanda Ekberg, Jonathan Eriksson and Gabriel Yousef. Your contributions made the project a success and we had a great time doing it.

We want to say thank you to Tommy Nordström at AFRY for the continuous support, quick problem solving and for introducing us to people that moved the project forward. To Patrik Holmberg, our supervisor at AFRY, for providing us with comprehensive information necessary to understand the subject and user needs, as well as for your valuable feedback and suggestions.

We would also like to send thanks to the people at AFRY, especially our supervisor Albin Möller and Oscar Bengtsson, for their wide number of suggestions which greatly improved the concepts. Their diverse perspectives and expertise provided us with new insights that steered the project in the right direction.

We express our gratitude to several individuals at KTH: Anna Hedlund Åström, for providing information regarding materials and supporting the thesis work; Claes Tisell, for your invaluable feedback on the concepts, suggestions for solutions, and assistance with calculations; Dr. Akepati Bhaskar Reddy, for your insights on stepper motors; and Mia Hesselgren, for your expertise in product branding.

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Stockholm, June 2024

TABLE OF CONTENTS

1 INTRODUCTION **1**
1.1 Background 1
1.2 Purpose 4
1.3 Delimitations 4
1.4 Method 4
1.5 Risk analysis 6
1.6 Planning 8
2 FRAME OF REFERENCE **11**
2.1 Market analysis 11
2.2 User study 13
2.3 Service outline 16
2.4 Brand Analysis 18
3 CONCEPT GENERATION **23**
3.1 Requirement specification 23
3.2 Quality Function Deployment 24
3.3 Concept ideation 24
3.4 Two concepts 2
3.5 Concept evaluation 32
4 CONCEPT DEVELOPMENT **35**
4.1 Detailed design 35
4.2 Material selection 48
4.3 Cost analysis 49
4.4 Prototype 53
5 RESULTS **55**
5.1 Conceptual results 55
5.2 Prototype 61
6 DISCUSSION AND CONCLUSION **65**
6.1 Discussion 65
6.2 Conclusions 68
7 FUTURE WORK **69**
7.1 Usability 69
7.2 Repairability 69
7.3 Design of the device 69
7.4 Service 71
8 REFERENCES **73**
APPENDIX A. WORK BREAKDOWN STRUCTURE **1**
APPENDIX B. GANTT-SCHEDULE **3**
APPENDIX C. INTERVIEW GUIDE **5**

APPENDIX D. REQUIREMENT SPECIFICATION 7
APPENDIX E. QUALITY FUNCTION DEPLOYMENT 9
APPENDIX F. SKETCHES 11
APPENDIX G. MORPHOLOGICAL MATRIX 13
APPENDIX H. VARIATIONS 15
APPENDIX I. BILL OF MATERIALS 17

1 INTRODUCTION

In the introduction the background, purpose and delimitation of the project will be described. The method that will be used will be explained. The introduction will also contain the risk analysis and planning of the project.

1.1 Background

The master thesis was conducted together with AFRY and the project included the development of an optical scattering measurement device. Optical scattering measurement devices are used to measure light reflection and light scattering from materials, to obtain data of the surface and bulk properties of materials. The measurement data are often used in research and development projects where material requirements are important, also for quality control in manufacturing processes, in different optical simulations and can be used for photorealistic rendering. Some examples of industries using scattering measurement data are paper production, medical technology, painting and interior design.

Light scattering measurement is the measurement of reflected or transmitted light from a material surface. The intensity of the optical scattering is dependent on the angle of incidence, wavelength, polarization and observation angle (Synopsys 2024). Light interaction with material surfaces is explained by scattering theories, and can be represented by the bidirectional scattering distribution function (BSDF) which contains the measurement data. When light propagates from a surface it is divided into reflected and transmitted light, where bidirectional transmittance distribution function (BTDF) and bidirectional reflectance distribution function (BRDF) is used accordingly. The BTDF and the BRDF consist of both specular and diffuse light, where the specular transmittance or reflectance happens on smooth surfaces and have the same angle of propagation as the incident light angle, which can be seen in Figure 1. The diffuse light is the ray scattering with a lower intensity that is affected by the material surface roughness. The result of the BSDF is the relation between the incident and reflected or transmitted light energy for a given number of angles (Ansys Optics 2024).

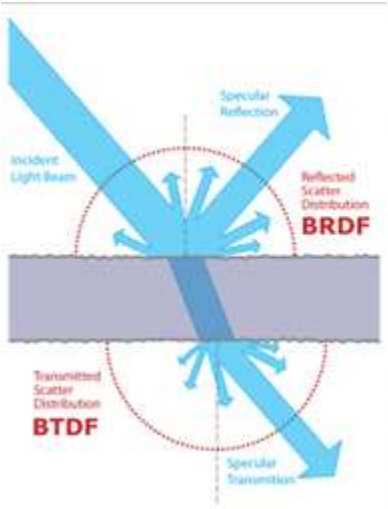


Figure 1. Visualization of BSDF (Ansys Optics 2024).

The specular light has an intensity significantly larger than the diffuse light, which in turn can have an effect on the end result of the measurement. It is difficult for the sensor to get correct data on the diffuse light scattering close to the specular light scattering, increasing the risk of an inaccurate result.

The optical system measuring the angular distribution and intensity of ray propagation, consists of a combination of a light source, a sensor, a sample, sample holder and an environment in which the measurements are conducted. The chosen optical system requires both pre- and post processing as a part of the applied model. Each subsystem has a number of variables to consider and affect the end result of the measurement.

Light source:

- Convergent or collimated distribution of light
- Mechanical stability
- Polarization
- IR to UV light
- Measurement of different angles of incident light

Sensor:

- Accuracy
- Range to consider
- Specular and diffuse light
- Motion for a spherical measurement
- Model type

Sample and sample holder:

- Material properties to consider
- Surface roughness
- Geometry
- Isotropic vs. anisotropic materials
- Stationary or dynamic sample holder
- Motion of sample holder

Environment:

- Non-reflective surroundings
- Color
- Material
- Temperature
- Humidity
- Stability of structure
- Hygienic

Pre Processing:

- Angular increments
- Stabilize motion
- Noise filtering
- Calibration
- Continuous processing
- Amplifier

Post Processing:

- Fitting of models
- Simulation
- Validation
- Model appropriateness for different light and material properties
- Data storage

The measurement devices that exist today do not collect the data optimally, do not concede for optimal use and are expensive to use. The technologies for measurement have different configuration, size and underlying measurement capabilities which affect the quality of data simulation.

1.2 Purpose

The purpose of the project was to develop an optical scattering measurement device for AFRY that can be used to measure surface and bulk scattering properties of materials to provide correct data for simulations. The goal was to develop a feasible concept and a functional prototype of the device that include the user interface both physically and digitally. AFRY wants the device implemented in a future service. The project had a time constraint of 20 weeks and a large scope, therefore a prioritization of functionalities was done to make sure the end product fulfilled the requirements. Further, the purpose of the final device was to create a prototype in order to validate the concept or parts of the concept.

The master thesis was conducted in a multifunctional team where the project members acted as mechanical engineers within mechanical design. The work was conducted together with masters students in software development and engineering physics with a focus on optics. The planning of the project was managed by the mechanical designers. It also included having an overview of the project to make sure the subsystems fit together.

1.3 Delimitations

The variables of each subsystem mentioned in section 1.1 will be divided in between the project members responsible for software, optics and mechanical design. In this master thesis the mechanical design will be accounted for and that is the environment, stability and movement of the optical components. Appropriate materials for the device will be selected with a focus on performance and sustainability. The device should meet the requirements while being sustainable. The sustainability will also be accounted for in the mechanical construction by considering repairability, recyclability, material hygiene and the implementation of the device in a service system.

The time constraint is a delimitation of the project and will be finalized in the beginning of June.

1.4 Method

The methods used to address the problem defined above will be explained in this chapter.

The project was carried out with the double diamond in mind which means that the project was conducted iteratively and has been process oriented. The double diamond approach is based on four stages of development, where the design project starts in the *Discover* phase, diverging into the *Define* phase. The *Develop* phase is diverging to visualize the investigation of different detailed design solutions. Lastly the *Deliver* phase is converging, it is the phase where the details of the project are finalized (Design Council 2024). The visualization of the double diamond process can be seen in Figure 2, where some of the methods used are presented.

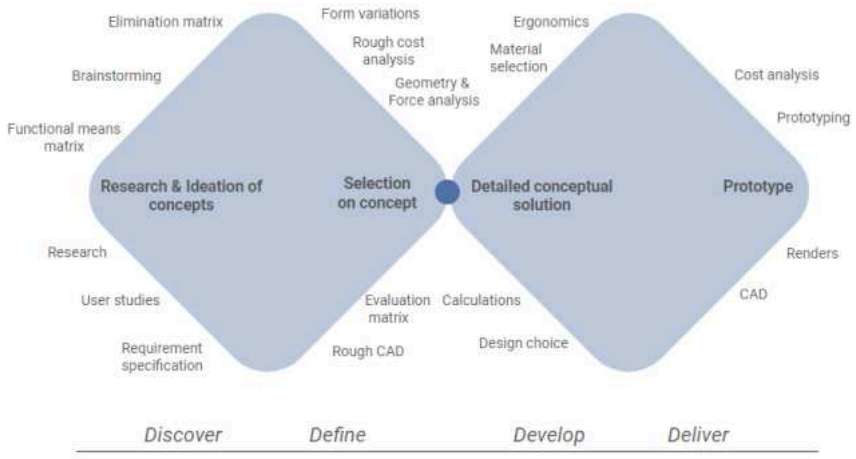


Figure 2. Visualization of double diamond methodology

The project was divided into six phases where each phase had its own deliverables. Phase 1 involved mostly mapping and planning of the project and a risk analysis was developed. Phase 2 consisted of literature study, user studies, material study, Quality Function Deployment, requirements specification and final mapping. Methods such as market analysis and brand analysis were used.

Morphological matrix, brainstorming, brainwriting 6-3-5, black box and elimination matrix are examples of methods that were used in Phase 3. These ensured that innovative ideas were discovered since several different methods were used and during different creative sessions.

In Phase 4 form variations and evaluation matrix were the chosen methods. Phase 4 also consisted of geometry analysis, force analysis, rough cost analysis and further ideation. The work was iteratively in terms of idea and concept generation, where value co-creation was used as a method as well. Value co-creation is to iteratively create the product together with the stakeholders and in that way create value more efficiently (Stickdorn et. al., 2018). In this project the value co-creation meant that the project members presented their concepts for the rest of the project, the supervisors and the users and considered the new input to further develop the concepts. In that way more information was collected before evaluating which concept to select and develop entering Phase 5.

In Phase 5 material selection, calculations, final component selection, design choices, CAD, renders and cost analysis were conducted. Phase 6 consisted of making the physical model as well as functional testing together with the final submissions.

The combination of the methods that were used throughout the project ensured that the end solution fulfilled the purpose.

1.5 Risk analysis

The risk analysis of the project can be seen in Table 1. Potential risks have been specified where each risk is graded on how likely they will occur and the consequence on the project if they would occur. The grading is 1, 3 and 9, where 1 is unlikely and 9 is highly likely. An action plan of the risks with the highest likelihood can be seen in this chapter too. Note that the identified risks regard the project as a whole and therefore some risks are not directly applicable to the mechanical design, for example the sensor calibration.

Table 1. Risk analysis

Risk	How likely it will occur	Consequence
The project member responsible for the optics is not finished with designing the optical system on time, which the rest of the project group is dependent on.	3	9
The sensor read the measurement data inaccurately.	3	3
There is a gap between the physical model and the conceptual model.	1	3
Wrong choice of resolution and method.	1	9
Wrongly selected requirements as a basis of decision making.	9	9
Faulty in sensor calibration.	9	9

Absence of project members.	1	3
The scope is too extensive in relation to the time limit.	1	3
Delay of tasks because of dependencies between tasks.	9	9

The action plan is as follows:

Wrongly selected requirements as a basis of decision making.

- Extensive literature and user study.
- Do not stress through the process of making the requirement specification. The process should include literature study, user study, research, market analysis, need analysis and functional analysis.
- Validate that the requirements are specified correctly and that no requirements are missing, through input from supervisors.

Faulty in sensor calibration.

- Test on samples where there are already existing measurement data.
- Plan for calibration when testing the prototype.
- Calibrate part systems before calibrating the whole optical system.

Delay of tasks because of dependencies between tasks.

- Regular meetings once a week where each team presents the previous week's work and the coming week's goals.
- Mapping the project through individual Gantt schedules, which are merged into a joint Gantt schedule.
- Ask each other, the supervisors and other experts to keep the momentum in the project.

1.6 Planning

In this section the planning of the project will be described. The first section explains the method used, called work breakdown structure (WBS). The second section shows the GANTT-schedule and the overall planning of the project.

Work Breakdown Structure

The WBS is used to structure the project and can be used as the first step of planning. The project work was divided into activities and milestones and structured both sequentially and concurrently. In multifunctional projects where competences overlap it is preferable to divide the responsibilities early and can be done at this stage. Dependencies between the areas of development are identified. Using the WBS helps the project group to estimate the length of each phase and cost of resources needed. Projects where the WBS is overlooked or not used are usually prolonged with higher costs and an affected workflow for the project members (Tonnquist 2021).

The project group conducted a workshop where the team members brainstormed on the activities to fulfill the deliverables within the project scope. The activities were divided into optics, software and mechanics. Activities dependent on each other were marked with arrows pointing in the direction of dependency. The WBS was used as the foundation for creating the GANTT-schedule, which will be presented below. The finalized WBS can be seen in Appendix A.

GANTT-schedule

The GANTT-schedule was used in order to create a detailed schedule of activities, estimated time and deadline of each activity. Both sequential and concurrent development were planned for. The sequential development is used where there are dependencies between activities and concurrent development is used when activities are performed simultaneously (Tonnquist 2021).

The GANTT-schedule for the project can be seen in Appendix B. Every task has a start date as well as an end date. In the scheme, public holidays and days where we are not available for work have been marked red and days marked orange means that the work needs to finish earlier that day.

The project was conducted during 20 weeks and was divided into six phases where each phase had its own deliverables. The deliverables can be seen in Table 2.

Table 2. Deliverables for each phase

Phase	**Deliverables**
Phase 1: Mapping and planning of project	Planning report, Gantt, risk analysis
Phase 2: Research and well defined design space	Requirement specification and final mapping

Phase 3: Ideation of concepts	2-3 feasible concepts
Phase 4: Selection of concept	Chosen concept
Phase 5: Detailed conceptual solution	Final concept including CAD and renderings
Phase 6: Prototype and final submissions	Physical model, final presentation, report and poster

In the beginning of a project it is easy and cheap to make changes while making changes late in the design process often leads to high costs. It is important to keep this in mind and especially in this project since it extends over a long period of time and implementing changes late can affect the end result. Therefore, extra time has been allocated in each phase to account for changes. Furthermore, certain parts of projects often take more time than originally expected. Consequently, the project was planned to be finished earlier than the official end date to make room for unexpected events and still meet the time constraint.

2 FRAME OF REFERENCE

Choosing between different types of optical scattering measurement devices, the project member responsible for the optical model of the device has selected a gonioreflectometer as a foundation for the development (Yousef 2024). The frame of reference will therefore include different aspects of that optical model. The market analysis will consist of which gonioreflectometers that is on the market today. Other devices relevant for the project will also be investigated in the market analysis.

The user study will be presented in this chapter and an analysis of the user needs. Since the device will be offered as a service, the product development will be affected and that will be explained in the section about the service outline. Further on, a brand analysis of AFRY will be conducted, where keywords that affect the formgiving of the product will be presented, which later on will be used in the concept generation.

2.1 Market analysis

The optical model that will act as a foundation for the development will be a gonioreflectometer. The gonioreflectometer consists of a light source, a detector for reflectance and transmittance, a sample area and a construction (source driver hoop and rotating annuli) that enables the desired motion. The different angles of reflection and transmission from the material sample will be detected sequentially and the measurement data will be collected. Figure 3 shows an example of a gonioreflectometer setup.

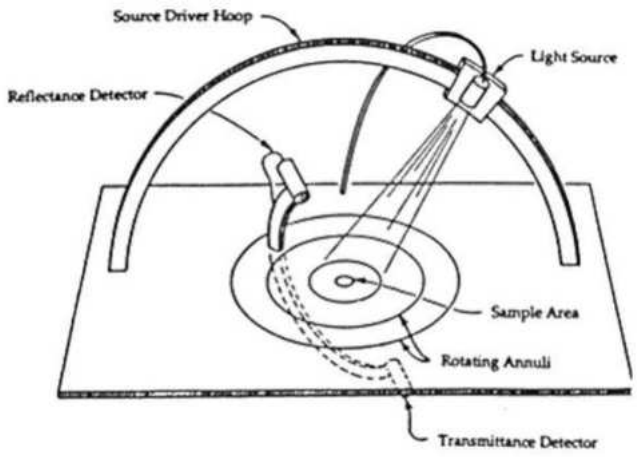


Figure 3. Gonioreflectometer (Yousef 2024)

The benefits of using a gonioreflectometer is a wide angular range with an angular resolution of less than 1 degree. The disadvantages are that it requires a complex mechanical construction with high demands of stability and accuracy. It can be time consuming if constructed similarly as the devices today, since the accuracy of the measurement increases with the number of collected data points (Yousef 2024).

Another type of a gonioreflectometer is the one seen in Figure 4. It is an automated gonioreflectometer that measures BRDF by using a robotic hemispherical motion. The detector is placed on the robot, the light source is seen to the right and the material sample is placed on the holder marked in red and white. The benefits of this type of a gonioreflectometer is the accuracy of the measurements that can detect a large variation of material and reflection properties. It can also account for surface roughness such as diffuse, shiny, specular, smooth, rough and partly transparent. It measures anisotropy and retroreflection in detail (Fraunhofer Institute of Optronics 2024). The disadvantage of this type is also the complexity of the construction, as well as the price of the components.



Figure 4. Automated gonioreflectometer using a robotic motion (Fraunhofer Institute of Optronics 2024)

An example of a goniophotometer manufactured by Realistic Graphics Lab (RGL) is shown in Figure 5. RGL uses a PAB pgII goniophotometer which means that the instrument facilitates concurrent illumination and observation of a vertically oriented sample from varied angles through four motorized axes. The illumination emanates from a stationary light source with a fixed distance to the material sample. The angle of incident light is controlled through the rotation of the sample mount. The surrounding environment is optimized for light absorption, by the application of black fabric to the room walls, mitigating unwanted reflection from the sample (RGL 2024).

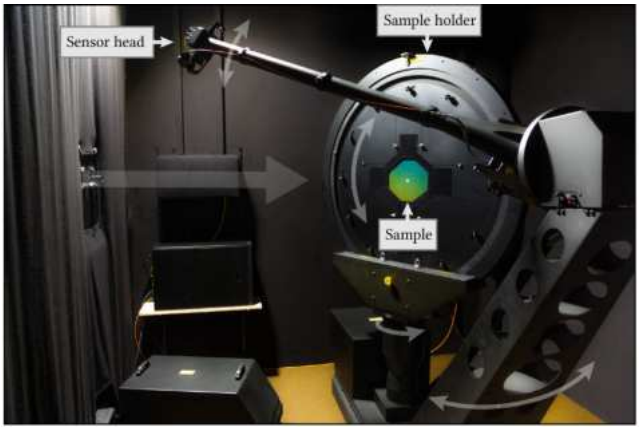


Figure 5. Goniophotometer BRDF/BTDF (RGL 2024)

An optical scattering measurement device requires a high pointing accuracy, which puts a large demand on the moving mechanism and its parts. Telescopes have similar demands of pointing accuracy, hence different solutions of moving mechanisms in telescopes will be investigated.

Celestron Telescopes are one of the most popular retailers for telescopes. They offer motors to automate the moving mechanisms in their telescopes. Some of the motor solutions on the market are the Rigelsys Stepper motor which has a 0.1 degree step and can lift up to 3.5 kilograms. Pegasus Astro Motor Focus Kit Zero uses a stepper motor with no backlash with a torque required for accurate focusing and repeatable movement. Universal Motor Focusing Unit is a high torque stepper motor that allows microstepping and high precision (Celestron 2024; PegasusAstro 2022; Astro-Gadget 2024).

2.2 User study

The user study consists of interviews and an analysis of the user needs.

2.2.1 Interviews

Five semi-structured in-depth interviews were conducted in total with the aim of discovering the needs of the user as well as to learn more about optical scattering. Four persons at AFRY were interviewed whereas all of them have expertise in optics and are the intended users. One interview was conducted with a scientist at Research Institute of Sweden (RISE), who has developed a product measuring light scattering for material simulations. The questions asked during the interviews can be reviewed in Appendix C.

The main insights from the interviews were the following:

Interface:

- Preferred interface is a smaller touchscreen outside of the device with the ability to connect your own laptop for more control.

Faulty messages:

- Users want a message on their phone where it is specified if something goes wrong.

- Green and red light on the device to indicate if it works correctly.
- Important to specify when there is a fault in the measurement.

Feedback:

- Confirm correct use with a symbol, for example a check box or similar.
- Confirm correct use on a low level, such as a flickering green light.
- Time estimate of measurement is important for the users and the ability to change the measurement settings to get the desired time estimate.
- Currently there is no feedback in similar devices and the user has to investigate why the measurement was not conducted as wished. The user would prefer interactive feedback and a step to step instruction of how to correct the faulty in the measurement.
- Resemble decoding.

Calibration:

- Reference samples, where the properties of the materials are known, can be used to calibrate the optical components.
- Relative measurement simplifies calibration.
- Prefer automated calibration of the device and that it occurs daily.

Measurement settings:

- The ability to choose between a default setting and advanced settings.
- Want to be able to quickly scan before a thorough measurement.
- The measurement settings from the previous measurement should be saved in a measurement log and allow the ability to select a previous measurement setting.
- The measurement settings should be printed on the same page as the measurement data.
- Different user levels, such as beginner, expert and manufacturer.
- One parameter should be able to be changed in a measurement setting, without changing the other parameters.
- Provide the ability to state expected intervals on measurement parameters and if the measurement data deviate from the interval, the user gets a warning message.
- Provide the ability to change parameters such as transmission or reflection measurement, angular increment of detector and light source, wave length, number of data points collected and start angle of incident light.

Other insights:

- Modularity is important and the ability to change optical components.

- Hygienic environment is important.
- Dust particles can affect measurement.
- Quick and easy to use without the need of instructions.
- Can repair if there is an instruction manual resembling IKEA step-by-step.
- The smaller the device the better.

2.2.2 RISE

The interview was conducted digitally on the 16th of February 2024 with the scientist Stefan Källberg working at RISE. The interview questions in Appendix C have been used as a base in the interview and Källberg also shared more information on his own work. Their group at RISE are doing measurements on optic scattering, through standard methods such as Haze (ASTM D-1003) and with their proprietary gonioscatterometer to be able to calculate quantities such as BTDF and BRDF. Källberg described his research and the technology used to be able to develop his own optical scattering measurement device.

The main insights from the interview were the following:

- Diffuse light can be measured with an angular resolution of 5°.
- Commercial measurement devices do not use standardized angular geometries, resolutions and use different mathematical models. In this way, the result of a measurement can differ from another, depending on which measurement device that has been used. It is not clearly defined and therefore difficult to interpret.
- Fiber collimator is used with the light source to collimate the beam and separate the beam from the surrounding environment, which gives a more homogenous incident light.
- The incident light beam is 20 mm in diameter.
- Uses a relative measurement by measuring the intensity of light on the sample in different angles in relation to the intensity of light on a reference.
- Uses a stepper motor and motor control for angular resolution and position.
- The size of the aperture in relation to the distance to the sample controls what angle is being measured. The smaller the aperture, the higher angular resolution. Commercial devices usually have one setting, which means losses in detail properties.
- Material samples with high diffuse reflectance and transmittance, require an angular resolution of 2-3°.
- It is desirable to have a high angular resolution and a small aperture, but the practical problem is that this setup results in a small intensity of reflected and transmitted light. The light propagates in all directions in space and when we want to detect a small part of that, the signal will drown in noise from the surrounding environment.
- Important to consider stray light, since it can disturb the signal.

- The detector needs to be linear in the area of measurement to be able to get a qualitative and absolute value.
- Angle of the incident light is controlled manually.
- Data collection and rotation of the detector is controlled automatically.
- A separating screen is placed between reflected and transmitted light to avoid contamination in the measurements.
- Every point of measurement takes 10 seconds.
- User interface for Källberg's gonioscatterometer is a computer.
- Desirable to do a quick scan of material to get a rough picture of what is being measured.
- Källberg wants different levels of user modes on a future device - beginner to expert.
- The evaluation of the measurements were difficult to conduct and as time consuming as the mechanical construction.

2.2.3 Analysis of needs

In order to get a better understanding of the needs of the user, an analysis of needs was made based on the conducted interviews. The physical, cognitive and emotional needs of the intended user was investigated in a methodological manner and is presented in Figure 6 (Ericsson et. al. 2017).



Figure 6. Physical, cognitive and emotional needs of the user

2.3 Service outline

In this project the optical scattering measurement device will be provided as a service. Product-as-a-service is a circular business strategy that is ecological and economical sustainable and can also strengthen social sustainability. Since there will be a shift from the product being used by a single customer to a plenitude of customers, it will put high demands on the product. The product development will go from a linear to a circular one. Using a circular business

model, such as Product-as-a-service keeps products on the market for a longer period of time, which produces more opportunities for stakeholders to create value from the product (Stena Circular Consulting 2022). See Figure 7 for a comparison of the linear and circular business models.

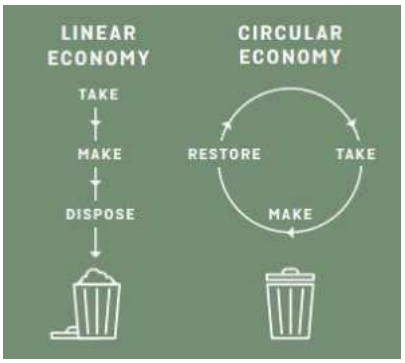


Figure 7. Linear and circular business models (Stena Circular Consulting 2024)

In accordance with the perspective of Walter Stahel, the Product-as-a-service model is the foundation in creating a resource-efficient circular economy, resulting in a diminished carbon footprint per product in correlation with number of users (Stahel 2010). The added value associated with the integration of a Product-as-a-service framework is documented as more personal customer relationships, sustained revenue streams and augmented competitiveness. Furthermore, the customers stand to benefit from cost reductions due to decreased product ownership coupled with access to cutting-edge technological advancements (Stena Circular Consulting 2022).

The implementation of Product-as-a-service imposes high demands on the product with respect to construction, material choices and usability. The construction and material choices need to be more sturdy to accommodate frequent use and users with varying levels of expertise. Given the diverse user base and the complexity of the product, focusing on usability is a high priority during the product development phase. The pivotal role of usability lies in repeat usage and the likelihood of recommending the service to others. The framework includes the creation of an extensive digital infrastructure that can handle maintenance, booking, payments as well as addressing the digital user experience (Stena Circular Consulting 2022).

Circular product design readiness stands for how easy a product can be implemented in a circular economy. Boorsma et al. (2022) advocate for the evaluation of circular product design readiness through a methodical examination of the product across various thematic domains and indicators. When developing the optical scattering measurement device and the possibility to provide it as a service, the theme of *customer experience* should be considered and that includes the product-use-efficiency. The product-use-efficiency has to do with on and off boarding of products between customers in the service as well as intuitive usage of the product. The second theme to consider is *product support service* which contains a professional support service including maintenance, repair and upgrades and a spare part supply. It is also essential to create a framework for recirculation of broken parts and design the product for disassembly, refurbishment, remanufacturing and recycling.

The optical scattering measurement device will be stationed at AFRY and the customers will be able to use the device in place as well as send their material sample to get a measurement conducted for them. The product will mostly be used by AFRY personnel, but the high demands of usability, construction and material choices are still relevant.

2.4 Brand Analysis

Since the project will result in a product from a brand that does not currently offer physical products, a brand analysis will be conducted. The brand analysis will help to frame the formgiving of the product and provide a more tangible description of the brand.

Firstly, the brand analysis will consist of investigating how AFRY communicates their brand. This will be interpreted through the mixer board tool as well as moodboards for the visual interpretation. Hopefully, leading to simplifying the idea and concept development of the project.

2.4.1 The AFRY brand

The AFRY brand will be analyzed based on the method visualized in Figure 8 below.

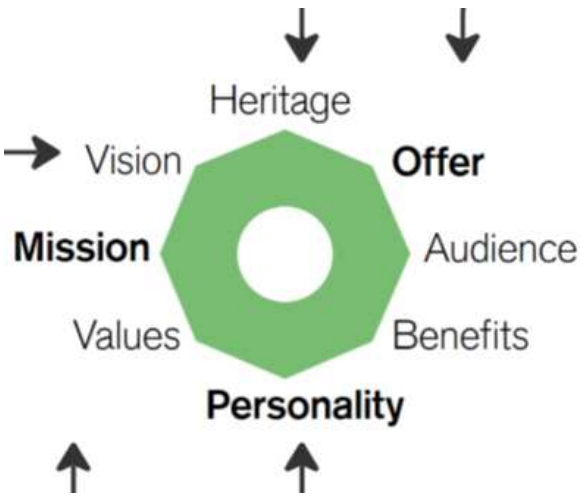


Figure 8. Brand analysis method (Hesselgren 2023)

Heritage:

AFRY was founded in 1895 in Sweden, with the name Ångpanneföreningen. Then the company offered development and repairs of boilers. Later expanding to include electrical engineering and other types of engineering services. It was renamed to ÅF Technology before changing to AFRY in 2019, through a merger between ÅF and the Finnish company Pöyry. AFRY is a global company with approximately 19 000 employees.

The business idea of AFRY is to accelerate the transition to a sustainable society, where the sustainability goals have a fundamental role in the business strategy. The goals include development of sustainable solutions and an ethical work environment.

The motivators of AFRY to drive the transition are explained as decarbonisation, circularity, electrification and digitalisation (AFRY 2024).

Offer:

AFRY's main offer is sustainable solutions.

Audience:

Customers in need of expert competence regarding engineering, design and advisory support.

Benefits:

Sustainable solutions developed by a diverse and global company.

Personality:

The AFRY personality is interpreted with the attributes professional, open minded, brave, challenging and innovative.

Values:

AFRY's values are brave, devoted and teamplayer. Brave is explained as thinking bigger and encouraging entrepreneurship. Devoted is characterized as a mix of competence, being passionate and curious to learn more. Teamplayer means sharing ideas, cooperation across borders and that diversity is a strength.

Mission:

AFRY's mission, which is the action based statement that fulfills the vision, is explained as accelerating the transition towards a sustainable society.

Vision:

AFRY's vision is making future (AFRY 2024).

2.4.2 Mixer board

To clearly visualize and communicate AFRY as a brand it will be interpreted through the The Designer's Mixer board tool, created by an industrial designer used in the course Advanced Product Design (Enlund 2023). A version of the mixer board can be seen in Figure 9. It consists of 9 rulers divided equally into Application, Semantics and Tweak. Application consists of *Surface, Direction* and *Mass.* Semantics consists of *Exhort, Describe* and *Express*. Tweak consists of *Hook, Wit* and *Dirt. Hook* can be explained as a design that draws the attention from the customer, *Wit* can be explained as a design that is humoristic and a handmade product often has more imperfections leading it to have high *Dirt*.

Since it is a professional product, no *wit* or *dirt* have been added. There should be some *hook,* since one of the values is brave and it should stand out. By expressing a lot of *mass* the device will feel trustworthy and stable. It should be clear how the product is supposed to be used, therefore the *describe*-ruler has a high value. It should *express* AFRY values, but without affecting the impression of being a lab equipment.

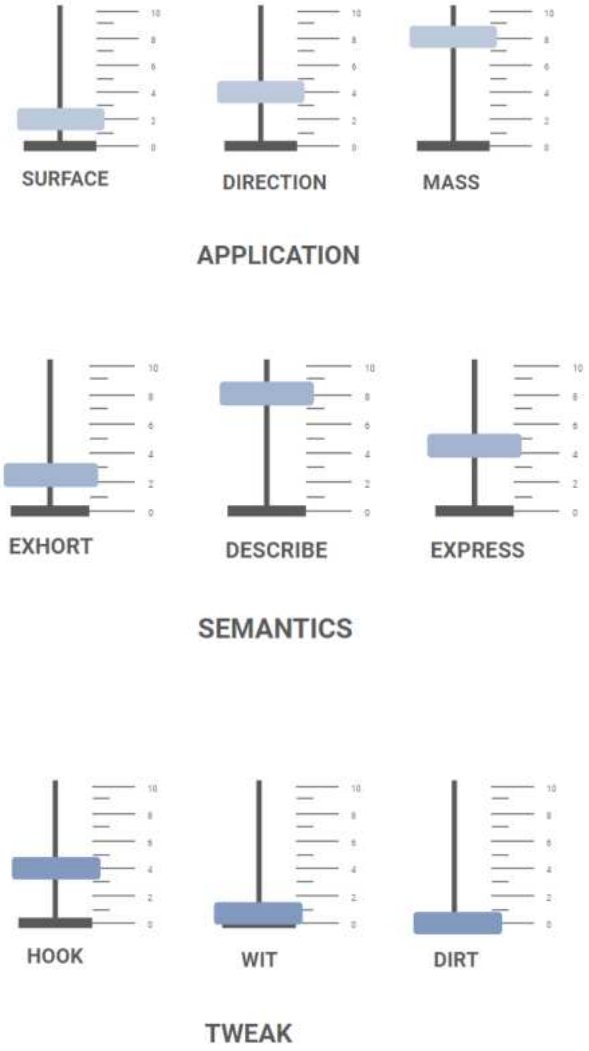


Figure 9. Mixer board rulers applied to Application, Semantics and Tweak

2.4.3 Moodboards

Moodboards were created in order to guide the design process (Ericson et al. 2017). The moodboards were conducted with *2.4.1 The AFRY brand* and *2.4.2 Mixer board* as a base for selecting suitable images and words. Figure 10 visualizes the moodboard regarding the formgiving of the device. The expression of the device is visualized in Figure 11.

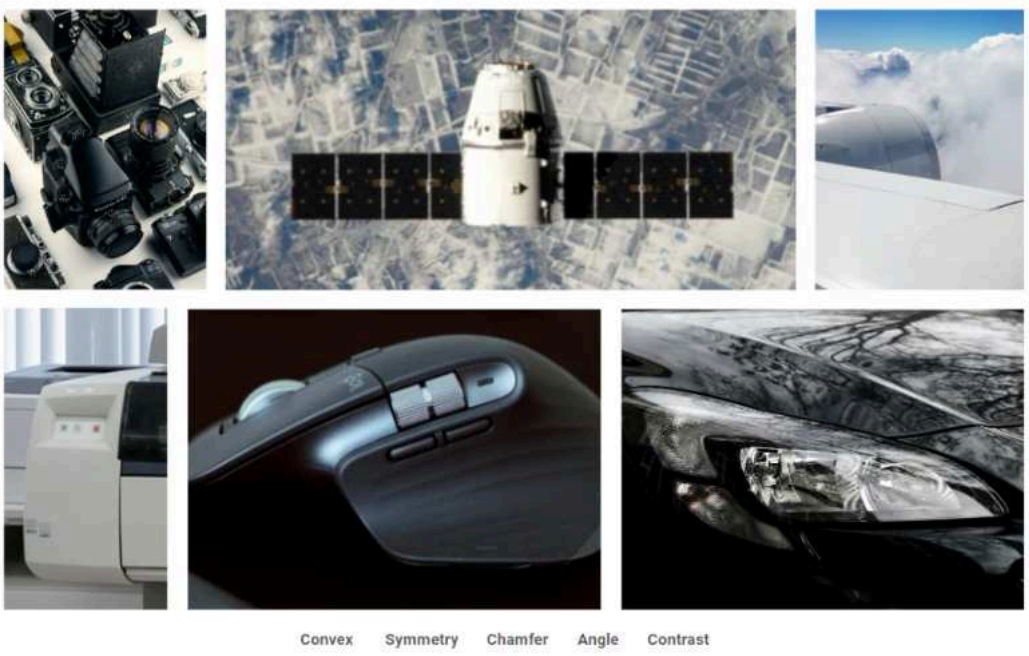


Figure 10. Moodboard formgiving

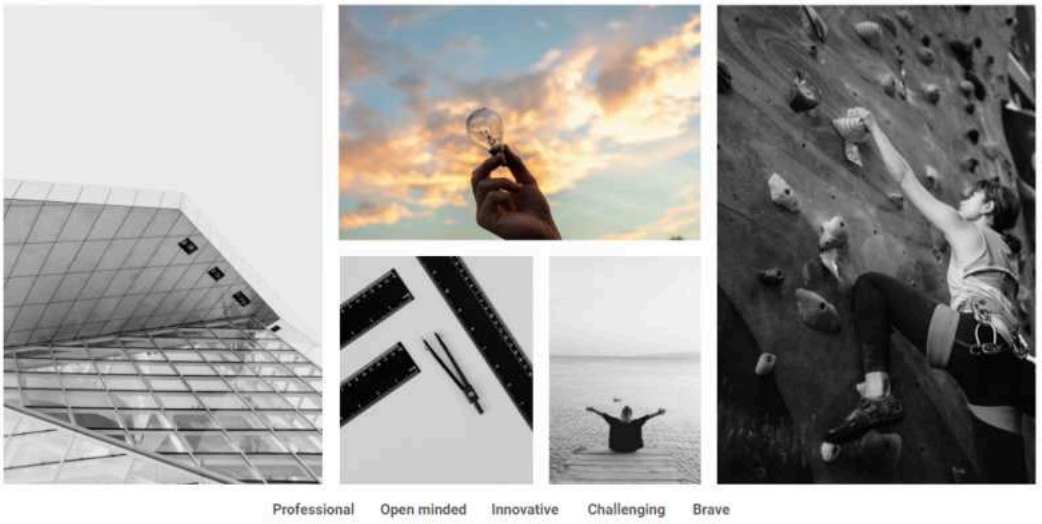


Figure 11. Moodboard expression

3 CONCEPT GENERATION

In this chapter the concept generation will be described. The requirement specification and Quality Function Deployment will be used as the frame of the concept generation and will be explained in this chapter. The ideas of the different areas to develop will be presented here as well as the two concepts. The concepts will be evaluated and one concept will be selected to develop further.

3.1 Requirement specification

The requirement specification was created with careful consideration to the research and interviews conducted. It serves as the frame of the project and a basis for idea and concept ideation. It has been divided into 1. *Main-* and 2. *Basic requirements, 3. Usability, 4. Ergonomics and Safety, 5. Sustainability, 6. Production and Mounting*, and *7. Market and Target group.* Each requirement is classified as either a requirement or a wish, which indicates what requirements to prioritize. The requirements have been specified through the method function analysis where the functionality of the product is defined without determining functional solutions (Ericson et al. 2017). Hence, the requirements will answer to what will be developed, not how it will be developed.

Examples from the requirement specification are the following:

- Enable spherical measurement
- Measurement of optical scattering, for specular and diffuse light
- Measure reflection and transmission
- Avoid intersection between detector and light source when measuring
- Allow movement of light source between measurements
- Allow for adaptable angular increment of light source and detector
- Fit different material samples
- Minimal human interaction before and after measurement
- Minimize risk of misuse
- Allow for different kinds of users
- Provide easy accessibility of sample holder between measurements
- Provide easy accessibility of light source and detector
- Be repairable
- Enable modularity
- Express AFRY values: brave, devoted and team player

The requirement specification can be seen as a whole in Appendix D.

3.2 Quality Function Deployment

A Quality Function Deployment (QFD) was conducted in order to transform the customer requirements from the conducted interviews to functional requirements. The QFD shows the relationships between the different requirements and states the importance of them (Ullman 2010).

The QFD for this project illustrated that the functional requirement *cost of production* is related to most customer requirements, such as modularity, easy to use and measure transmission/reflection. The cost of production increases with the complexity of the product, but it is important to keep the cost low. Since many customer requirements are related to the cost of production it will be prioritized in the project and will be used as a factor when making decisions.

The QFD can be seen as a whole in Appendix E.

3.3 Concept ideation

In this section the early concept and idea ideation will be explained. Brainstorming has been used and the ideas will be presented. An evaluation matrix will be used to select the spherical movement that satisfies the requirements the best.

3.3.1 Brainstorming

The method used for generating ideas was primarily brainstorming (Ericson et al. 2017). It was conducted in several sessions with different focus in each session, both individually and cooperative. The principles were discussed regarding the sample holder, spherical movement, environment and ways to access the sample. Further on, the principles of the different categories were visualized through sketching. The principles are explained below and for further understanding the sketches can be viewed in Appendix F.

Sample holder

- Wedged
- 2-point fastening
- 4-point fastening
- Laying flat
- Pocket
- Frame

Sample access

- Drawer
- Door
- Rotate around 1 point

- Click to open
- Bottom door

Spherical movement

- Globe vertical and horizontal
- Arm with 3° of freedom
- Cage with rail tracks

Environment

- Box
- Sphere
- Organic shape
- Cupboard
- Cube
- Cylinder

3.3.2 Elimination matrix

An elimination matrix was used in order to early spot the ideas that do not meet the requirement specification and should therefore not be continued working on (Ericson et al. 2017). The focus was primarily on finding the desirable spherical movement, since it affects which sample holder to be selected, how to access the sample and which environment it should operate in. The elimination matrix can be viewed in Table 3. The matrix shows that the spherical movement should be fulfilled by the Globe principle.

Table 3. Elimination matrix of spherical movement.

Requirements / Ideas	Globe	Moving arm	Cage
Enable measurement of reflection and transmission	+	+	+

Detector needs no repositioning	+	-	+
One intersection between light source and detector during measurement	+	+	-
No movement of sample holder	+	+	+
Minimal material needed for construction	+	-	-
Design enables reparability	+	-	-
Design enables modularity	+	-	-
Feasible construction	+	-	-
Allow for adaptable angular increments of optical components	+	+	+
Time efficient	+	+	-
Adapt for different users	+	-	-
Feasible within time frame of project	+	-	-
Continuation of work	Yes	No	No

3.3.3 Morphological matrix

After selecting the globe as the desirable spherical movement, the other parts of the device will be selected. In order to generate a lot of different solutions of each part, a morphological matrix was made. The morphological matrix includes criterias, in the left column, that explain what should be developed. The matrix is built up with how each criteria can be developed. These are combined differently to create concepts (Ritchey 2002). The two concepts that were selected are presented in the next section.

The morphological matrix can be seen as a whole in Appendix G.

3.4 Two concepts

The two concepts with the highest potential for future work were chosen. The principle of movement, sample holder and separating screen are similar in both concepts and will be described in detail below. The concepts are in desktop size, both user friendly in terms of ergonomics, safety and intuitive use. Both concepts have potential to be implemented in a service because of the high usability.

The concepts are called 1 and 2 and are visualized below. Formvariations of the outer shell of both concepts and drawer for the material sample can be seen in Appendix H. From the form variations, the visualization of the two concepts were created.

3.4.1 Principle of movement

The two concepts share the same principle of movement where the material sample is fixed and the detector moves along a rail in 0-180° in the polar direction (θ), simultaneously as the rail with the detector rotates 0-360° in the azimuthal direction (φ). A light source can move along a different rail between 0-85° in polar direction while the rail stays fixed. The detector has an angular step of 2° and the light source has a 5° angular step, according to the user study. See Figure 12 for a visualization.

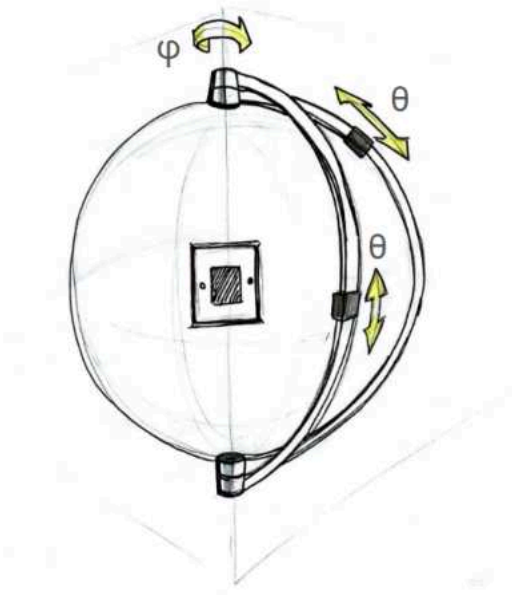


Figure 12. Principle of spherical movement

3.4.2 Sample holder & separating screen

With inspiration from the RISE interview, a separating screen was chosen for both concepts in order to avoid light contamination between reflection and transmission measurements.

The material sample holder is visualized in Figure 13. The material sample will be placed between two frames that will in turn be placed inside of the separating screen. The frames are constructed as passepartouts so that there is an opening in the center where the light source can reach the sample and the two frames are held together with knobs or screws. To assemble the material sample the whole sample holder is derived from the separating screen and the material sample is then placed between the frames, outside of the device. This limits the human interaction needed with the device and there is therefore less risk of accidentally misaligning objects when mounting the material sample.

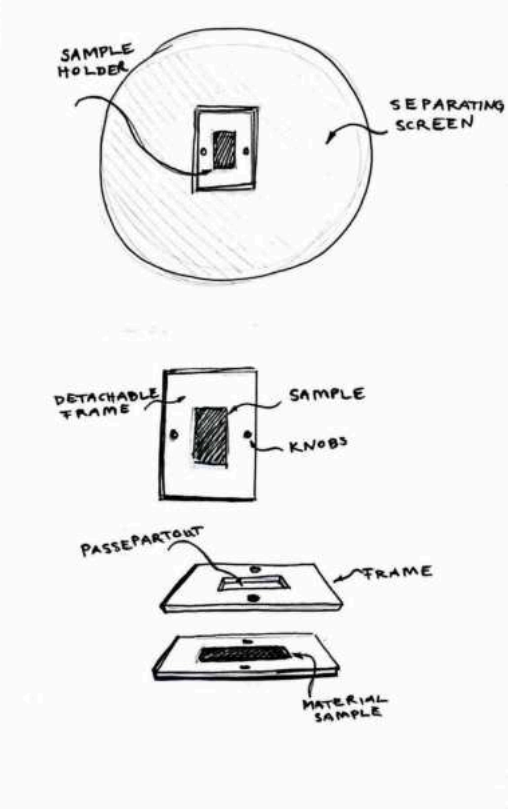


Figure 13. Early sketches of sample holder and separating screen

3.4.3 Concept 1

Concept 1 uses the principle of movement explained in *section 3.4.1* and can be seen in Figure 14. The material sample is vertically oriented and placed in the separating screen. To place the sample holder into the separating screen, the user opens the front door and places it in the separating screen with a click. The user chooses appropriate settings before starting the measurement. After the user has conducted a measurement the light source and detector moves

to their starting positions, at the top of the device. The top door is used to access the detector and light source in case of exchange or repairs. The outer shell has the purpose of making the environment dark in which the measurement is conducted. It also protects the optical components from dust. The formgiving of the external design has been created with the brand analysis as a basis. It has a lot of mass to visualize trustworthiness and stability. It is symmetric, professional and resembles the shape of a satellite, which is one hook to the design.

Variations made of the shape can be seen in Appendix H.

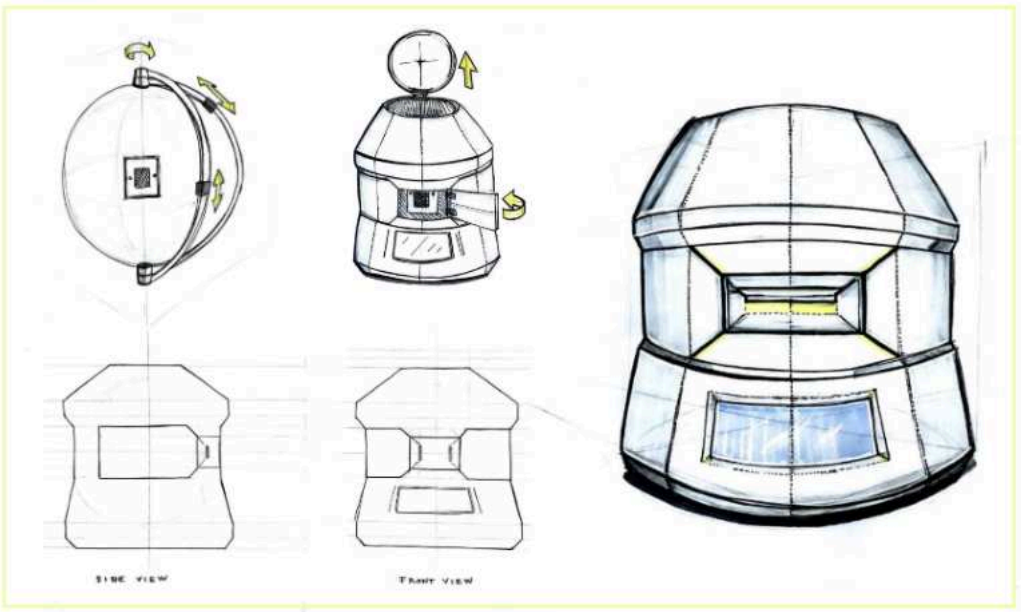


Figure 14. Sketches of Concept 1

3.4.4 Concept 2

Concept 2 uses the same principle of movement but is placed horizontal. It has the same separating screen and sample holder. To place the sample holder into the separating screen, the user pulls the handle and places it inside. This limits the human interaction further and is also more ergonomic since it is seen from above and there is no need to hunch down in order to place the material sample. This concept also allows for easy repairs since it is easy to add repair hatches on top as well as on the side to access components. See Figure 15 for a visualization of the concept. Variations made of the shape can be seen in Appendix H. The outer design of the device is not decided upon and what is visualized is more of a base for future design work. The base is visualized with the brand analysis in mind and Concept 2 has also a lot of mass to visualize stability and trustworthiness. The formgiving is convex with chamfers to express a professional feeling, resembling the formgiving of a lab equipment.

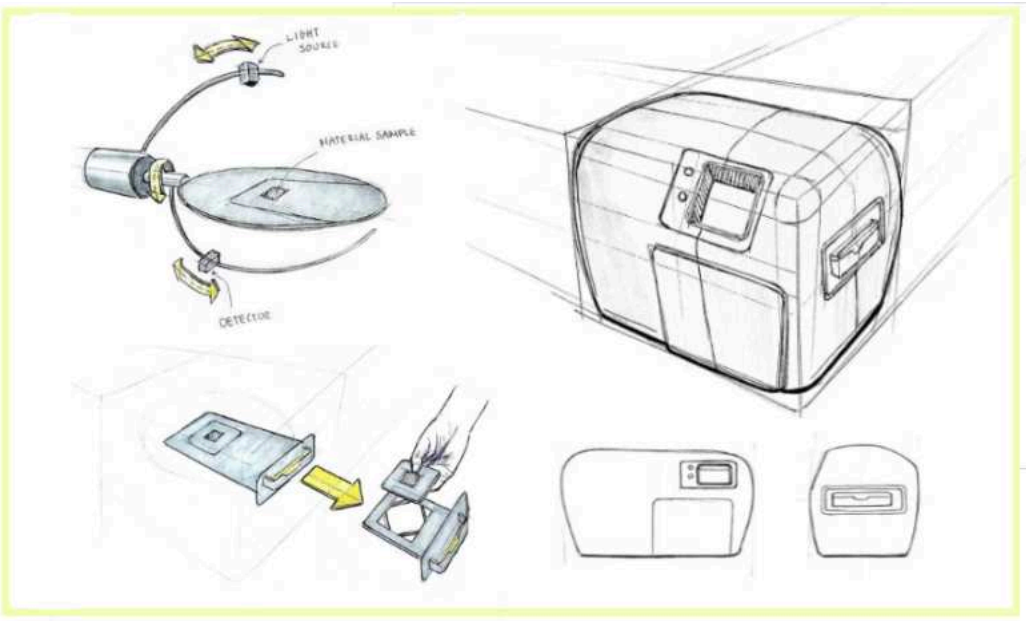


Figure 15. Sketches of Concept 2

3.4.5 Early calculations

The movement of the detector and light source of the two concepts can be enabled by using a motor that moves along on the rails. This means that the motor also needs to move itself, which can be heavy and difficult to handle the mounting of the motor together with the optical components. Therefore, it is more desirable to have the motor stationary outside of the movement. This can be achieved by using gears to move the optical components. One pinion gear attached to the motor axle and one large gear to enable the correct motion track of the optical components, which is illustrated with a small and large circle in Figure 16 as well as Figure 17.

Before evaluating the two concepts, calculations will be made of the torque needed by the motor to move the detector on its track.

Concept 1
To calculate the torque needed by the motor a schematic image was made which can be seen in Figure 16.

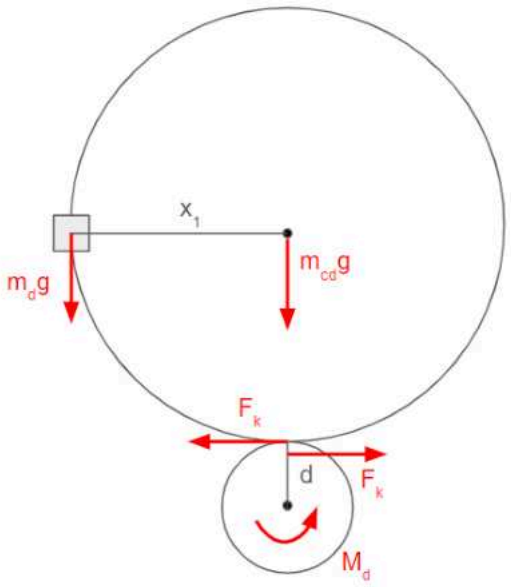


Figure 16. Concept 1

Since the center of mass of the fixating structure is in the same line as the torque M_d, it will not be accounted for. The torque needed by the motor can be calculated as

$$\circlearrowleft: \; m_d g \cdot x_1 - F_k \cdot x_1 = 0$$

$$\circlearrowleft: \; M_d - F_k \cdot d = 0$$

where the distance is x [mm] and the weight of the detector is m_d [kg].

Estimations were done of the gear force F_k to account for the weight of the detector as well regarding the distance d, which resulted in an estimated torque for Concept 1 to M_d = 0,7 Nm.

Concept 2

The torque needed to drive the motor for Concept 2 can be calculated in a similar way to Concept 1, see Figure 17.

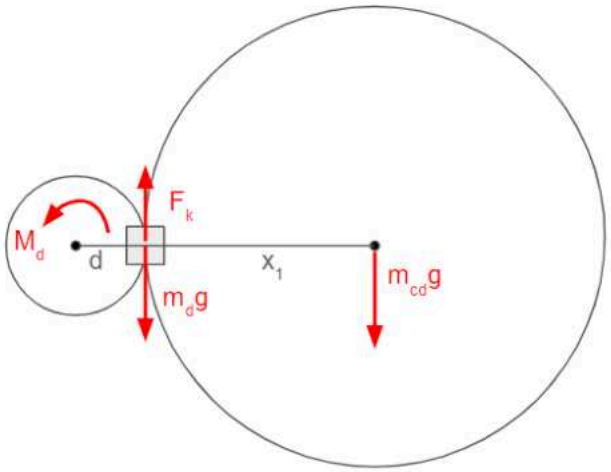


Figure 17. Concept 2

For Concept 2 the torque M_d needed by the motor is calculated as

$$\circlearrowleft: \; m_d g \cdot x_1 - F_k \cdot x_1 = 0$$

$$\circlearrowleft: \; M_d - F_k \cdot d = 0$$

which results in the same torque needed as Concept 1.

3.5 Concept evaluation

The two concepts are evaluated using a weighted criteria matrix where each criteria is weighted and the concepts are graded on every criteria (Ericson et al. 2017). It can be seen in Table 4.

Table 4. Weighted criteria matrix

Criteria	**Weight**	**Concept 1**		**Concept 2**	
		Score	Total	Score	Total
Low torque needed by motor	5	4	20	4	20
Low cost	3	5	15	3	9

Easy construction	5	4	20	2	10
Minimize human error	3	3	9	5	15
Total			64		54

Concept 1 and Concept 2 both have a low torque needed by the motor. Concept 1 got the highest score and has a less complex construction than Concept 2. Concept 1 has a lower expected price than Concept 2. Concept 2 got a better score on minimizing human error because of the drawer solution accessing the material sample. Since Concept 1 has got a better score on most of the criterias as well as the highest total score, the chosen concept is Concept 1. The chosen concept will be developed further in detail.

4 CONCEPT DEVELOPMENT

The concept development will primarily describe how to obtain the movement of the detector and light source. It will present the detailed design and each component required for the movement. The motor needed for the movement will be selected based on calculations described here. The material sample holder and the separating screen will be presented. Suitable material and a cost analysis of the device will be explained. A physical prototype will be created to validate the movement, since the movement of the detector and light source will be similar, only the detector movement will be prototyped. The prototype will be presented in this chapter.

4.1 Detailed design

The detailed design of the concept has been iterative. A lot of solutions have been investigated and presented to the team members, supervisors and users, and feedback has been received. In that way the concept has been co-created with the involved parties by considering their feedback and suggestions. The solutions presented in this chapter is the final selected concept for this project. The dimensions of each component are presented in section *5.1 Conceptual results*. The prototype will need some alterations that will be explained in chapter *7. Future work.*

In Figure 18 the concept and its parts are presented. The main components are shown and stated in the Figure.



Figure 18. The detailed concept and its parts

In Figure 18 the bearings allow movement in the azimuthal direction. The internal gears allow movement in the polar direction for the detector and light source. The separating screen separates the transmission and reflection measurements from each other to avoid light contamination. The light source is on the outer movement track and the detector on the inner movement track. The material sample holder is in the center of the separating screen.

4.1.1 Motor selection

When selecting suitable motors for the device a number of requirements were taken into consideration. It is desirable to have the same motors in the whole device. The motors should be able to hold torque, perform at low speed, provide large accelerations and retardations, have a high angular accuracy, hence a high control. Telescopes have similar motor requirements and as can be seen in the section *2.1 Market analysis*, which uses stepper motors provide these results. Stepper motors have a relatively low cost which is desirable in this project. Consequently, suitable stepper motors will be selected for the device.

When selecting stepper motors, it is based on the movement with the largest loads and torques in the system. The conducted calculations are the forces and angular accelerations of the detector movement in both polar (θ) and azimuthal (φ) directions. In Figure 19 a schematic sketch of the forces and torque of the detector system is presented, as well as arrows to show the polar and azimuthal directions.

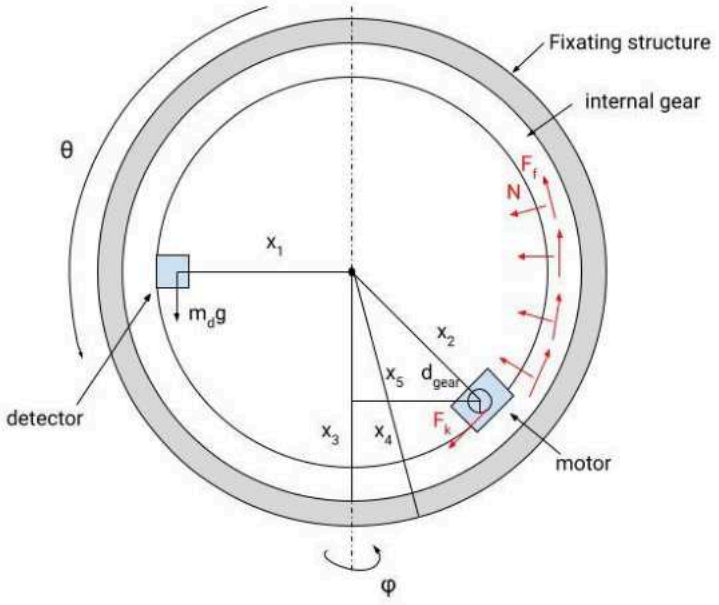


Figure 19. A schematic sketch of the forces and torque of the system

Figure 19 shows the supporting construction which is static, the internal gear moving counter clockwise, the detector that is attached to the internal gear and is rotated by a gear attached to the motor which enables the movement by providing a torque.

Force analysis θ-direction

The force analysis in the θ-direction is conducted to investigate what gear force F_k and normal force N that the motor can overcome and select an appropriate motor. The friction force F_f [N] is

$$F_f = \mu N \tag{1}$$

where the coefficient of μ is estimated to 0.4. The required torque M [Nm] is

$$M = m_d g \cdot x_1 + F_f \cdot x_3 \tag{2}$$

where the weight of the detector m_d is 0.3 kg, the distance from the center of rotation to the detector x_1 is 200 mm and the radius of the internal gear x_3 is 200 mm. The torque M [Nm] is also represented by

$$M = F_k \cdot x_2 \tag{3}$$

where F_k [N] is the gear force and the distance from center of rotation to the center of the gear x_2 is 188 mm. The motor holding torque M_θ [Nm] is

$$M_\theta = F_k \cdot \frac{d_{gear}}{2} \tag{4}$$

where the diameter of the pinion gear d_{gear} is 150 mm (Apazidis 2019).

The holding torques of the motors were chosen based on the calculations of M_d from *section 3.4.5* together with two different safety factors. The first alternative of a motor was one with a holding torque of 1.8 Nm and the other option was 1.2 Nm. Different safety factors were tested in order to investigate the selection of an appropriate motor.

Using the equation (1), (2), (3) and (4) combined, and a motor with a holding torque of 1.8 Nm, resulted in F_k = 24 N and N = 53 N. A motor with a holding torque of 1.2 Nm resulted in F_k = 16 N and N = 33 N.

In this stage of the calculation, some of the values selected are estimates as a starting point of the calculation. The equations are iterated on with the final selected values and measurements to verify that the values of the gear- and normal forces do not exceed the initial predicted values.

Force analysis φ-direction

In the azimuthal direction it is important to consider the angular acceleration of the system to be able to know what motor is needed to overcome it.

The torque M_{φ} [Nm] in the azimuthal direction is

$$M_{\varphi} = J \cdot \frac{\Delta\omega}{\Delta t} \tag{5}$$

where J [kg·m^2] is the moment of inertia and $\frac{\Delta\omega}{\Delta t}$ [rad/s^2] is the angular acceleration. The moment of inertia J_{ring} [kg·m^2] of the supporting construction is

$$J_{ring} = \frac{m_{ring} \cdot x_4^2}{4} \tag{6}$$

where the mass m_{ring} is estimated to 1 kg and the radius x_4 is 250 mm. The moment of inertia in the azimuthal direction is dependent on both the inertia in the polar and azimuthal direction which is

$$J = m_d \cdot x_1^2 + J_{ring} + m_m \cdot x_5^2 \tag{7}$$

where the mass of the detector m_d is 0.3 kg and the distance from the center of rotation to the detector x_1 is 200 mm. The mass of the motor m_m is estimated to 1 kg and the horizontal distance from the center of rotation to the motor x_5 is 100 mm.

Using equation 5, 6 and 7 combined, the angular acceleration can be solved for (Apazidis 2019).

A motor with a holding torque of 1.8 Nm resulted in an angular acceleration $\frac{\Delta\omega}{\Delta t}$ = 47.8 rad/s^2. A motor with a holding torque of 1.2 Nm provide the angular acceleration $\frac{\Delta\omega}{\Delta t}$ = 31.9 rad/s^2.

The force analysis in the θ-direction is the one that is crucial for dimensioning and selecting the accurate motor. The parameters to consider for the motor selection are the ones calculated above for the polar and azimuthal directions, which resulted in the selection of a stepper motor with a holding torque of 1.2 Nm and an angular step of 1.8°, encoder, motor driving unit and power supply, which can be seen in section *5.1.2 Components.*

The purpose of the encoder is to accurately position the detector. The encoder tells the motor driving unit where the stepper motor is positioned. This ensures the accurate angular position of the detector, which is of high importance during the data collection.

4.1.2 Gear ratio

The step angle for the selected stepper motor is 1.8°, therefore gears can help to translate into the correct angular step for the output gear. The detector should move with an angular step of minimal 2° and the light source should move with an angular step of minimal 5°.

The user also wants to be able to select how many data points to collect during a measurement, which means that the motor driving system needs to be flexible. The default angular step in the final device will be as mentioned, but the mechanics should still provide the flexibility the user needs.

This is solved by creating the proper gear ratio in the system. Having a gear ratio that obtains a 0.1° step of the gear will provide the desired flexibility in the system. This means that in the software that controls the motor, a code snippet needs to be inserted and it should send 20 pulses to the motor in order for the gear to move 2°. The step of 0.1° allows the user to have different angular steps of the detector and light source without the need of rebuilding the mechanical design.

To obtain the desired angular step of 0.1° of the output gear, the gear sizes, module, number of gear teeth and gear ratio is solved for. The gear ratio *i* is

$$i = \frac{angular\ step\ in}{angular\ step\ out} \tag{8}$$

The gear ratio is also explained by

$$i = \frac{d_2}{d_1} = \frac{z_2}{z_1} \tag{9}$$

where d_i [mm] is the diameter of the gears and z_i is the number of gear teeth of each gear. The pitch diameter *d* [mm] is

$$d = z \cdot m \tag{10}$$

where *m* [mm] is the module of the gear (KTH 2008).

This resulted in a gear ratio of 18. To avoid undercut of the gear a minimal number of teeth was chosen to 17. The pinion was therefore chosen to 17 teeth and the internal gear was consequently 306 teeth. Given from the requirement specification, the distance from the material sample to the detector should be 200 mm, this in turn means that the internal gear for the detector should have a pitch diameter of around 400 mm. To achieve this, the module was therefore calculated to 1.3. The light source internal gear uses the same gear ratio and therefore has the same amounts of teeths but has a module of 2.4 to achieve a larger pitch diameter.

4.1.3 Movement of detector

In spherical coordinates, the movement of the detector will go from 0-180° in the polar (θ) direction and 0-360° in the azimuthal (φ) direction. It is enabled by the following components.

θ-direction



Figure 20. Components that enable movement of detector in θ-direction

In Figure 20 the components that enable movement of the detector in θ-direction are presented. The internal gear has a width of 5 mm and an outer diameter of 420 mm. There are gear teeth 200° in the internal gear, since the movement of the detector is from 0-180° and the pinion gear and motor need a place to operate without disturbing the detector movement.

The internal gear has a mounting solution for the detector interface with two M5-holes in it for attachment. The detector interface holds the detector in place and has an offset of to compensate for the offset that is needed to have a fixating structure rotating around the separating screen. The offset of the detector holder enables the detector to point against origo accurately.

The detector holder is fitted to the dimensions of the detector and because of a delay of the information regarding the dimensions of it, which resulted in an intersection of the bearing that holds the supporting fixating structure and the detector interface. To avoid the intersection with the gears and axle the movement of the detector will decrease to 151°. The holder was not iteratively developed due to the time constraint and therefore an arbitrary design was made. The detector interface can be seen in Figure 21.



Figure 21. Detector interface

The supporting circular construction to the internal gear has an outer diameter of 432 mm and an internal diameter of 410 mm. The supporting circular construction is divided in two separate parts with different cross-sections to simplify the assembly process. The cross-sections of the different parts can be seen in Figure 22.

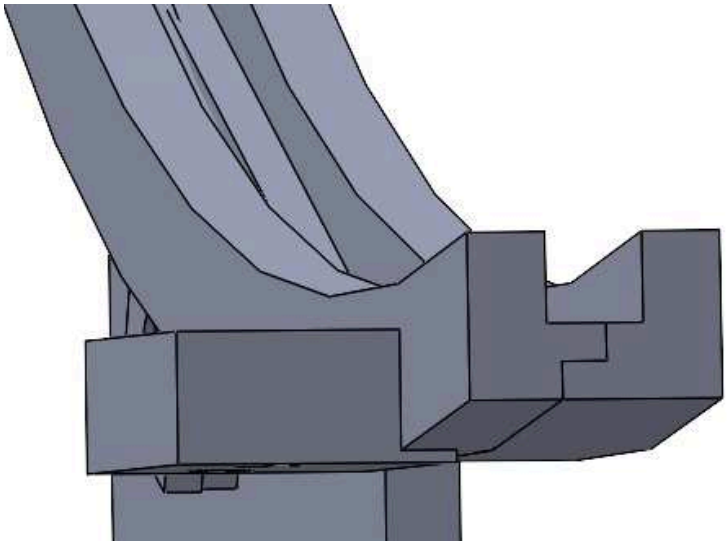


Figure 22. Cross-section of the two parts of the supporting circular construction

The two parts are fastened together with a click and with the use of M5 fasteners it is secured even further.

The supporting circular construction has a motor platform attached to it. It also has a horizontal fastening solution with M5 holes on it, which will be fastened to the bearing underneath. The circular construction is shifted to the side of the main axle of the construction, this is due to separating screen and associating material samples needed for support. Since the detector needs to be able to move 360°, an axis concentric above it would hinder the movement. The shift can be seen in Figure 23.

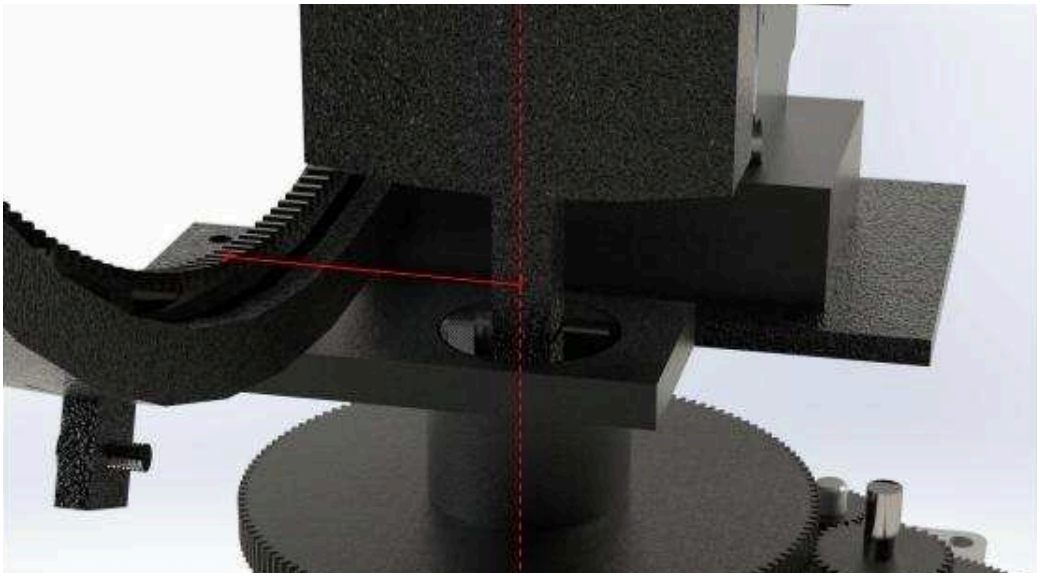


Figure 23. Shift to side of axle

The thickness of the circular construction is dependent on the light source angular step of 5 degrees, since a requirement of the device is to minimize blindspots. The maximum thickness is therefore calculated by using the equation of a circular arc, which can be seen in equation 11.

$$b = r \cdot 2\pi \cdot \frac{\alpha}{360} \tag{11}$$

The variable b [mm] is the maximum thickness and the length of the circular arc, α [°] is the angular step of the light source which is α = 5° and r [mm] is the distance from the material sample to the detector, which is r = 200 mm. With the stated values, the maximum thickness of the supporting circular construction of the detector is approximately 17.5 mm but was chosen to 17 mm to have a small margin.

φ-direction

In Figure 24 the bottom components that enable the movement of the detector in φ-direction are presented. The components that enable the φ-movement of the detector are a top bearing, a bottom bearing with a gear, a combination of gears that enables the desired angular step and a motor and gear platform.

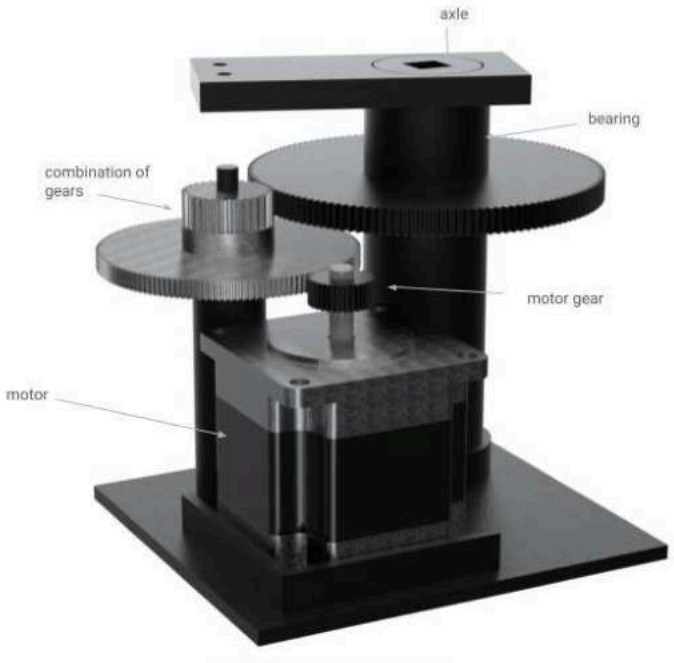


Figure 24. Bottom components that enable movement of detector in φ-direction

The top bearing has a horizontal plate with M5 holes to fasten the circular construction. The height of the top bearing is adjusted to position the circular construction to enable the detector to point to the origin of the sphere.

The bottom bearing with the gear also has a horizontal plate that enables the circular construction to be fastened. The bottom bearing has a gear welded on it. The bearing gear is connected to the combination of gears, where bearing gear acts as the last output gear. The bearing gear and the combination of gears makes the stepper motor that rotates with a 1.8 degree step to provide an output of 0.1 degree step for the detector, providing the mechanical system with the flexibility needed. The detector angular step will be 2° for this device, but it is easy to adjust it to other angular steps by changing the software to send other numbers of pulses to the motor. For instance, if the desired angular step for the detector would change to 1 degree, the software can be changed to send 10 pulses to the motor instead of 20.

For achieving the same gear ratio as the internal gear, two sets of gears are being used. This is conducted in order to make the pitch diameter smaller in relation to their positions. This is calculated through the equation for several gears (KTH 2008). The two sets of gears have the same ratio but different modules, this because they needed to be different sizes to not collide with the axle. Equation 12 shows the total gear ratio u_{tot} which is dependent on the gear ratio of the two sets of gears u_1 and u_2.

$$u_{tot} = u_1 \cdot u_2 \tag{12}$$

The motor and gear platform holds the gears and the weight of the motor. A cutout in the platform is created to provide space for the encoder and its axle.

4.1.4 Movement of light source

The movement of the light source was initially stated to go from 0-85 ° in polar direction, however the supervisor later had a wish for the light source to move in the whole sphere. Therefore the mechanical system was created to allow that movement. The light source movement will be 0-180 ° in the θ-direction and 0-360 ° in the φ-direction.

To simplify the manufacturing process and create a unified design, the components for the light source movement are the same design as the components for the detector movement. The difference is the dimensions, which will be presented below.

θ-direction
In Figure 25 the components that enable the movement of the light source in θ-direction are presented. The gears are a different material than the rest of the components in the figure to make them distinguishable for explanation purposes. The internal gear has a mounting solution for the light source interface with two M5-holes in it. The purpose of the light source interface is to attach the light source holder. The fixating structure for the light source does not have the same restriction of dimensions as for the detector, therefore it has a thickness of 10 mm to provide more stability. The motor platform for the light source has similar dimensions as for the detector.



Figure 25. Components that enable movement of light source in θ-direction

φ-direction
The components that enable the movement of the light source in the azimuthal direction are

similar as the components for the detector, and can therefore be seen in Figure 24 and is explained in section *4.1.3 Gear ratio*. The dimensions of the gears are the same as for the detector, the dimensions can be seen in section *5.1.2 Components*.

4.1.5 Axle parts

The axle of the device is divided into different parts to simplify the assembly process and to ensure the right positioning of the supporting circular construction, the material sample holder and the bearings. In Figure 26 the axle parts can be seen.



Figure 26. Axle parts

The axle parts have a diameter of 30 mm. Every part has either a plug or a hole, to stabilize and fasten the parts in each other. The plugs and holes have a square cross-section, the plug 10x10 mm and a length of 25 mm and the hole has the same length but has dimensions of 10.5x10.5 mm. The square cross-section is selected to disable rotation of the axle parts. See Figure 27 for a close up at the axle connection.



Figure 27. Close up axle connection

4.1.6 Material sample holder

The material sample holder is explained and visualized in section *3.4.2 Sample holder & separating screen*. The fasteners of the material sample holder are countersunk in the frames to avoid undesirable reflection. The fasteners are M5 screws and nuts. The nuts are placed inside of a fitted hole in the material sample holder. Different lengths of screws can be used to allow for different sizes of material sample thickness. Figure 28 shows the material sample holder with the maximum allowed thickness of 10 mm for the material sample.



Figure 28. Material sample holder in profile

When the light source beam hits the material sample there is a risk that the sample will reflect light onto the holder if there is an edge. To solve this, the holder has a tilt around the opening to the material sample, see Figure 29.



Figure 29. Material sample holder in perspective

The frames of the sample holder have round openings that have a diameter of 60 mm. This means that the material sample needs to be greater than 60x60 mm. The opening is larger than intended, this is due to the size of the beam from the light source. Since the light source will sometimes be pointing from the side, it will intersect with the material sample holder and it needs to be large enough to pass through the sample on both sides. Additionally the beam will not go straight through the sample due to refraction. See Figure 30 for a visualization of the beam pointing from the side, where only the transmitted light is shown for explanatory purposes.



Figure 30. Visualization of light beam hitting material sample from the side with transmitted light

4.1.7 Separating screen

The separating screen is explained and visualized in section *3.4.2 Sample holder & separating screen*. The separating screen is shaped to enable the placement of the motor and gears in the fixating structure. The separating screen has room for the material sample holder. It can be seen in Figure 31.



Figure 31. Separating screen

The separating screen has a diameter of 365 mm and a thickness of 20 mm. The thickness is adapted to be the same as the sample holder, with the material sample's maximum thickness placed inside. There is a cut-out that matches the sample holder, the protruding part out the sample holder cut-out is hard to tell in Figure 31 as it is not completely cut through. Due to the cutout, it is only possible to place the sample holder in the separating screen from one way. There is a square cross-section of the plugs that are assembled in the connecting axle parts, with the same dimensions as explained in section *4.1.5 Axle parts*.

4.2 Material selection

Selecting suitable materials for the components in the optical scattering measurement device is of importance when it comes to optimal performance of the mechanical system as well as the optical system. The optical system requires a low level of reflectance in the material of the surrounding components to avoid contaminating the measurement data of the material sample. The mechanical system needs to be designed to enable movement of the optical components, to hold the loads in the system and provide stability in the dynamic and static components.

The azimuthal rotation of the optical components are conducted by using bearings and to obtain the rotation in the bearing they need to have low friction.

The loads in the system that need to be considered are calculated in *section 3.4.5* and *4.1.1* and reach a value of 0.6 Nm and 40 N. The detector and light source will have weights of 0.2 kg and 0.3 kg respectively. The motor itself has a weight of 0.7 kg.

A meeting with Anna Hedlund Åström, the associate professor in material science at KTH Royal Institute of Technology, was conducted 2024-05-07. Questions were asked regarding material recommendations of different components. From the meeting it was recommended for the gears to have heat treated steel to make it stronger, endure abrasion and with high surface strength. The internal gear is advised to be in steel together with lubricant to enable a smoother movement. The supporting circular construction and the axle parts are suggested to be in surface treated aluminum, since aluminum is light-weight. More investigation is needed before selecting materials.

When selecting materials for the mechanical construction it is essential to also consider the aspect of ecological sustainability. This can be obtained by implementing repairability, recyclability and material hygiene in the device. The recyclability of the device increases with higher material hygiene, while the repairability has more to do with the formgiving and design of the components. When the device is enacted as a product-as-a-service, there will be more frequent use of it, hence a more robust material is needed. Therefore, a material is needed that fulfills the requirements stated above, that can be used in as many components as possible and meets the need of a robust material.

Black anodized aluminum is a multi-treated aluminum that enhances the properties of the material. The treatment makes it stronger and more robust and is widely used in aerospace, military sector and automotive industries. It lasts up to 20 years depending on the use frequency and what dye and treatment details that have been used.

The benefits of black anodized aluminum are the increase of corrosion permanence and durability against wear. An electrochemical treatment creates an oxide layer on the aluminum surface ensuring prolonged functionality and longevity.

There are different types of anodizing aluminum that give it different properties. Type II utilizes sulfuric acid to facilitate the formation of a denser and more adaptable oxide layer, increasing the endurance against corrosion and wear. Type II is known for its flexibility as well as low cost and is applied in a spectrum of applications ranging from consumer products to architectural designs (RallyPrecision 2023).

The treatment is made in different steps and for this device it is recommended to pre-treat it with a method called etching. Etching makes the aluminum matte, by adding heated sodium hydroxide. It removes a layer of the material, which is what creates the matte finish (AAC 2024).

The anodizing has the potential to diminishing surface friction, which is advantageous where moving parts interact, causing smoother operations and mitigating wear over time. The black oxide layer can also affect the heat emissivity of the aluminum surface which is a good choice for components requiring efficient heat dissipation. The black oxide layer makes the aluminum material to facilitate more effective heat radiation into the surrounding environment. Furthermore, black anodizing serves as the final touch to elevate the quality and visuals of the design of a product.

On the moving parts, such as the internal gears and bearings, there is a need for even lower friction and higher durability. Using a technique called PTFE-impregnated anodizing of the aluminum where polytetrafluoroethylene is assimilated into the anodized layer makes the surface lubricous and highly durable. It is especially advantageous for gears and hydraulic cylinders and a good choice for the optical scattering measurement device (RallyPrecision 2024).

The chosen material for the static components in the mechanical construction are black anodized aluminum and for the movable parts in the device a tradeoff between sustainability and performance needs to be conducted. PTFE-impregnated anodized aluminum is recommended for its ability to make the surface lubricous and durable, however it has a negative effect on the environment. For sustainability aspects, heat treated steel is a better alternative for the movable components.

The aluminum is recyclable and easy to downcycle. It has an average price of 48 SEK/kg (Ansys Granta EduPack 2024).

4.3 Cost analysis

In this part of the project it is in its place to calculate the material and manufacture cost of the device. The 1-3-9 methodology is used to calculate the manufacturing cost, which means that the estimated manufacturing cost will be three times the material cost of each component. The 1-3-9 methodology is often used when making products that intend to be produced at a large scale (Ullman 2010). Therefore using this method for one product will act more as a rough estimate to make sure it is not too expensive for further development. With the selected materials, the cost of the materials and the weight of each component and following costs are calculated and presented in Table 6, 7 and 8.

Table 6. Cost analysis of components enabling movement of detector

Component	**Weight [kg]**	**Material**	**Cost of material [SEK/kg]**	**Material cost [SEK]**	**Manufacture cost [SEK]**

Detector holder	0.047	Black anodized aluminum	48	2.26	6.78
Internal gear	0.080	PTFE-impregnated black anodized aluminum	48	3.84	11.52
Pinion gear	0.002	PTFE-impregnated black anodized aluminum	48	0.10	0.30
Supporting circular construction	0.271	PTFE-impregnated black anodized aluminum	48	13.00	39.00
Bearing bottom with gear	0.088	PTFE-impregnated black anodized aluminum	48	4.22	12.66
Bearing top	0.042	PTFE-impregnated black anodized aluminum	48	2.02	6.06
Double gear	0.026	PTFE-impregnated black anodized aluminum	48	1.25	3.75
Motor gear	0.002	PTFE-impregnated black anodized aluminum	48	0.10	0.30

Table 7. Cost analysis of components enabling movement of light source

Component	Weight [kg]	Material	Cost of material [SEK/kg]	Material cost [SEK]	Manufacture cost [SEK]
Light source holder	0.047	Black anodized aluminum	48	2.26	6.78
Internal gear	0.456	PTFE-impregnated black anodized aluminum	48	21.89	65.67
Pinion gear	0.002	PTFE-impregnated black anodized aluminum	48	0.10	0.30
Supporting circular construction	0.820	PTFE-impregnated black anodized aluminum	48	39.36	118.08
Bearing bottom with gear	0.088	PTFE-impregnated black anodized aluminum	48	4.22	12.66
Bearing top	0.042	PTFE-impregnated black anodized aluminum	48	2.02	6.06
Double gear	0.026	PTFE-impregnated black anodized aluminum	48	1.25	3.75
Motor gear	0.002	PTFE-impregnated black anodized aluminum	48	0.10	0.30

Table 8. Cost analysis of axle parts, separating screen and material sample holder

Component	**Weight [kg]**	**Material**	**Cost of material [SEK/kg]**	**Material cost [SEK]**	**Manufacture cost[SEK]**
Axle bottom light	0.154	Black anodized aluminum	48	7.39	22.17
Axle bottom detector	0.154	Black anodized aluminum	48	7.39	22.17
Separating screen	1.575	Black anodized aluminum	48	75.60	226.8
Material sample holder	0.141	Black anodized aluminum	48	6.77	20.31
Axle top detector	0.080	Black anodized aluminum	48	3.84	11.52
Axle top light	0.314	Black anodized aluminum	48	15.07	45.21

The estimated material cost of the components will be 124 SEK and the manufacture cost 372 SEK, which in turn results in a total cost of 496 SEK. Apart from the costs mentioned, the price of manufacturing labor needs to be considered.

4.4 Prototype

The purpose of the prototype is to validate the movement of the detector. Since the light source has a similar movement as the detector and because of the large dimensions of the parts associated with the light source, it was decided that proving the movement of the detector is sufficient.

Before the final prototype was made, several tests on smaller parts were 3D printed. This was made to see if certain movements and parts fit the way they intended to do. Parts that were printed in small scale were an early idea of the fixating structure and internal gear as well as axle parts with plug and hole connection.

As mentioned, the size of the finished product and the size of regular 3D printers are not compatible. This has been challenging as it has led to difficulties in printing the fixating structure of the detector. Dividing it into smaller parts is one option but that means that the final part will be less durable, and the fixating structure has high demands on durability and stability as it holds the detector and internal gear as well as is connected to the bearings.

Laser cutting the fixating structure in whole parts will create more stability, but that means the motor platform and bearing connections need to be 3D printed and attached to the fixating structure afterwards. Creating the fixating structure through laser cutting was the selected manufacturing method.

5 RESULTS

In the results chapter the conceptual results includes renderings of the optical scattering measurement device, dimensions of parts, choice of motors and a total cost of the device is presented. The final result of the prototype is shown here.

5.1 Conceptual results

In this chapter the overall device will be explained and the components as well as the movement will be presented in detail.

5.1.1 Overall

The optical scattering measurement device has a total height of 1052 mm and a width of 792 mm. Including the light source will make the total width larger although that is dependent on the placement of the light source versus the light source holder, to which there were no given recommendations. An overall view of the device can be seen in Figure 32. In Figure 33 the light source is aimed from behind. Figure 34 shows the device from the front. Figure 35 shows the device with an incident light beam.



Figure 32. Overview of the optical scattering measurement device



Figure 33. The optical scattering measurement device with the light source aiming from behind

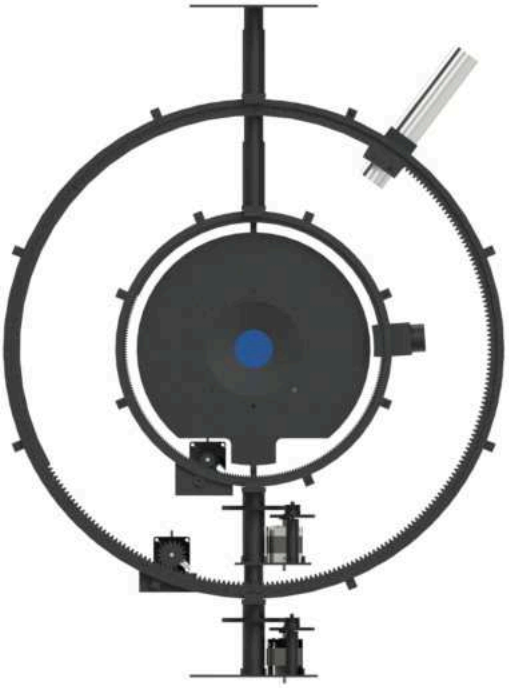


Figure 34. Device seen from the front



Figure 35. Device with an incident light beam.

The scattering device can measure with the detector in a range of 151°. This has to do with the range and placement of the detector and future work is needed to fulfill the desired movement, see chapter *7. Future work* for further explanation.

The detector can move with the angular step of 2° and the light source can move with the angular step of 5°. It measures in both BRDF and BTDF. Blind spots where the detector can not detect light will occur when the detector passes the light source and when the detector passes the separating screen as it gets blocked by the thickness of the screen.

The time for one measurement is an estimation based on the time it takes for the detector to collect data from one measurement point and the amount of data points that will be collected.

See Table 9 for information regarding estimated time for one measurement and thickness as well as dimensions of the material sample.

Table 9. Measurement information

Estimated time for one measurement [hours]	1.5

Maximum thickness of material sample [mm]	10
Allowed dimensions of the material sample [mm]	60x60 < X ≤115x115

The cost of the aluminum components is 496 SEK and for one motor and associated parts the cost is 3482, which results in a total cost of the mechanical system for the optical scattering measurement device at 14 424 SEK.

In Appendix I all parts of the components are presented as a bill of materials.

5.1.2 Components

Below are the choice of motor and gears presented, the rest of the components are described in chapter *4.1 Detailed design*.

Motor
Table 10 presents the chosen motor and the components associated with it, all which are from the retailer OEM Motor. A model of the motor has been downloaded from OEM Motors webpage and used in the CAD and renderings (OEM Motor 2024).

Table 10. Choice of motors and associated parts

Motor	Fulling stepper motor FL57STH56-2804B-UEN
Motor driving unit	Motor driving unit from Leadshine DM556E
Power supply	1-phase Crouzet power supply
Encoder	E5 512 CPR

Gears
The chosen gears and their properties can be seen in Table 11. The colors indicate that the gears are in the same pair and therefore have the same module.

Table 11. Teeth, module, pitch diameters and quantity of the gears used in the product.

	Teeth	Module	Pitch diameter [mm]	Thickness [mm]	Quantity
Internal gear detector	306	1,3	397,8	5	1
Pinion gear detector	17	1,3	22,1	7	1
Internal gear light source	306	2,4	734,4	10	1
Pinion gear light source	17	2,4	40,8	12	1
Gear motor azimuthal	140	0,5	70	5	2
Pinion motor azimuthal	33	0,5	16,5	7	2
Gear bearing azimuthal	140	0,7	98	7	2
Pinion bearing azimuthal	33	0,7	23,1	9	2

5.1.3 Movement

The movement of the detector in the θ-direction can be seen in Figure 36 and the movement of the detector in the φ-direction can be seen in Figure 37. This represents how the detector will move during one measurement while the light source stays in the same position.



Figure 36. Movement of detector in the θ-direction



Figure 37. Movement of detector in φ-direction

Figure 38 presents how the light source can change position in the θ-direction and Figure 39 presents how the light source can change position in the φ-direction. The repositioning of the light source will occur before measurements take place.



Figure 38. Movement of light source in θ-direction



Figure 39. Movement of light source in φ-direction

5.2 Prototype

The prototype is made at a 1:1 scale. Several changes have been made of the concept in order to be able to prototype it. For example, the axle parts have been divided into smaller parts in order to fit in the 3D printers. Parts have been redesigned to be able to be printed. Only focusing on the detector movement means that some axle parts have been remade to act as supports at the top and bottom instead of the light source axle components. In the prototype the material sample holder is missing, since the purpose of the prototype was to validate the movement of the detector. The internal gear for the detector as well as the fixating structure has been laser cut out of acrylic plastic and the rest of the components have been 3D printed in PLA. The parts have been spray painted in a black mat color to mimic the final lock of the device and also for reflective purposes. The components of the prototype can be seen in Figure 40. The assembled prototype can be seen in Figure 41 and 42.

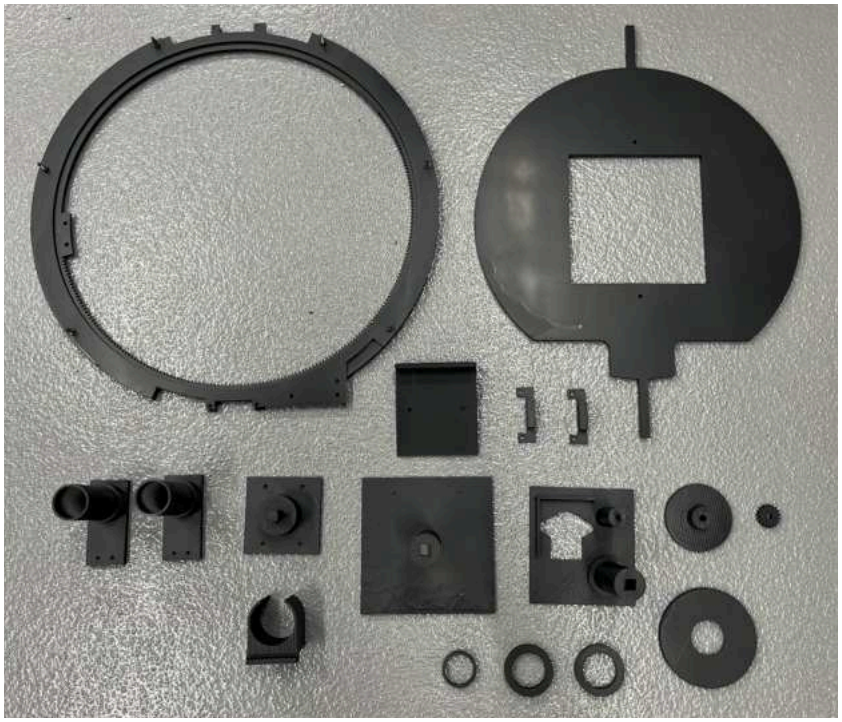


Figure 40. Components for the prototype



Figure 41. Assembled prototype front view

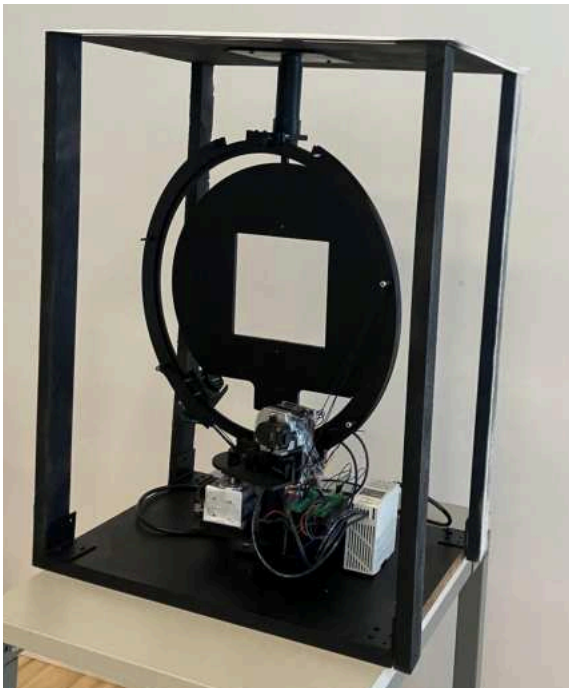


Figure 42. Assembled prototype

The detector was able to move in the polar direction with the help of the gears and motor and the fixating structure moved in the azimuthal direction with the help of motor and respective gears. A few quick alterations were made to fulfill the movement, for example fastening the motors and a way of hinderings the pinion gear on the motor to move sideways on the internal gear. The full movement of a sphere was not tested since the cords to the motor associated components were too short to allow that.

6 DISCUSSION AND CONCLUSION

In this section the discussion and conclusions of the project are explained.

6.1 Discussion

The purpose and goals of the project was to develop an optical scattering measurement device, where the focus for us within mechanical design were the environment where the measurement is conducted, the stability and movement of the optical components and selecting appropriate materials regarding performance and sustainability. In the mechanical construction we aimed to take sustainability into consideration by making it repairable, recyclable, with high material hygiene and do research on how to implement it in a service system. We aimed to create a conceptual solution and a functional prototype to test the movement of the optical components.

The purpose and goals of the project were partially reached, the goals that were not reached will be presented in chapter 7 about future work. A conceptual solution of the device was created that fulfilled the requirements of the stability and movement of the optical components as well as the environment in which the measurement is conducted. The materials selected took both performance into consideration by using black anodized aluminum which has low reflectance, high durability, light-weight and with the PTFE-treatment that made it lubricant and enabled movement on the moving parts. The sustainability was taken into consideration in the material selection by having high material hygiene and selecting materials that maintain the material properties over time. However, a tradeoff between sustainability and performance was made regarding the moving parts. To make it even more sustainable it is recommended to use heat treated steel in the moving parts.

The components are held together either with screws, fitting plugs or click, and glue is avoided when assembling. This means that the components are easy to disassemble in case of repairs and recycling, which is beneficial from a sustainability perspective.

The environment in which the measurement is conducted, was developed and presented as a sketch in chapter 3.4. We chose to delimit the scope further when we realized the complexity of the internal construction of the moving components, hence the external design was not developed. The user and brand study gave us a lot of insight on how it can be developed in terms of usability, ergonomics and formgiving and should be further developed with these insights in mind.

In the beginning of the project we put a lot of effort into planning the whole project and highlighting dependencies between the project members. We set dates for when the optical system needed to be selected and when the requirement specification needed to be completed. The dates of completion were agreed between the project members and presented to the supervisor. Having wrongly selected requirements as a basis for decisions was also stated in the risk analysis as a potential risk. But regardless of this, crucial information that affected the requirement specification and the framework of the project came up as late as the beginning of May. This gave us little to no time to make alterations of the design, which affected the end result in a negative way. One example was the dimensions of the detector. The moving track and mounting solution was designed with the original dimensions, but with the new dimensions the range of movement was affected. Another example was the requirements regarding the motion of the light source, which was initially set to 0-85 ° in polar direction. In the middle of the project it

was extended to 0-180° in polar direction and 0-360 ° in azimuthal direction, which is a comprehensive change. The conceptual solution fulfilled the light source movement, but that meant it was less time for changes in the CAD and the prototype. This event is called scope creep where the scope is made broader without adjusting the time or budget in proportion to the added or altered functionalities, which in turn affects the end result (Tonnquist 2021). To avoid scope creep, we argue that the requirement specification cannot be altered after a specific date without changing the time limit of the project.

As for now, AFRY does not have any products on the market which means that developing a product for them includes analyzing and interpreting their brand. Making a brand analysis is a basis and frame of a product development and especially if they want the product in a service. Making a product that only focuses on the functionality without considering the brand, would not be beneficial in a service. In this project we made a brand analysis, mood boards and mixer boards as well as a service outline. We would have wanted more time to iterate on the formgiving of the product and be able to visualize it in CAD to validate its suitability to the AFRY brand. We recommend to use the brand analysis as a basis for further work on the formgiving of the external design, and use the sketches we have made as a start of that process.

The user study included in-depth interviews with people that have used similar products or have an expertise in the area. The interviews gave us a lot of useful information and was used as a base for the requirement specification. But the interviews were conducted for each area of the project - the software and user interface, the optics and the mechanical design. Instead we would suggest that the project members responsible for the software and optics would conduct their own interviews regarding their subject. This would have given us more time to conduct interviews with a focus on the mechanics, the movement, the material sample holder and the usability of the mechanical system. If we would have had time for more interviews, that would have meant more data to strengthen the decisions of the mechanical design and less risk of inaccurate conclusions.

The detailed design and construction were more complex than expected and there were no straightforward solutions to implement the concept ideas. Because of the time limitation, the methodology used for each component was not as structured as the methodology used for the product concept development. Each sub-system and component design of the concept should have been developed iteratively and tested as well as evaluated in its context. A suggestion of further work is to use the same structured methodology used for the concepts as for each sub-system and component of the product.

The users wanted the device in a portable size, which can be made by decreasing the distance between the detector and material sample as well as between the light source and material sample. But decreasing the distance also means making the thickness of the fixating structure of the detector smaller, because of the blind spot requirement. Therefore, it is a balance of stability and reliability of the construction and of having a small and portable product. To know the minimal thickness that keeps the construction stable for a smaller product, solid mechanics calculations need to be conducted. As for now, the maximum thickness is calculated from the angular step requirement and set as a basis for further refinements.

It was challenging to be in a master thesis project, where there were many dependencies in between project members with other areas of expertise. In retrospect, it was difficult to know how late alterations would affect other project members' work and in what ways we worked differently. It was not optimal to work in parallel with all the areas in the project, and we assert that the optic part of the master thesis project should have been before we had ours. In that way the optical system should have been in place when we and the software team started our master

thesis project and we would have avoided a number of delays. In hindsight, the extensive size of the project scope and goals should have had a longer time frame than 4.5 months or should have been divided into several master thesis projects that fulfilled part time goals of the large project.

The motor selection was based on the calculations made in *section 4.1.1* which is intended for a single product. If the product will be produced on a larger scale it is recommended to do the same calculations but with the final values of the device. This will ensure the motor to not be overdimensioned which can increase the costs as well as the weight.

The dimensions of the device made it difficult to create a prototype. This due to the size of most 3D-printers, leading to constraints in what can be created in one piece. Bigger 3D printers used in the industries often have lower accuracy than the smaller ones. The prototype in this project had high demands of accuracy of certain parts and therefore had to be divided into smaller pieces in order to be printed. This has led to more uncertainty of the stability of the prototype and is also very time-consuming. Ideally, the prototype should have been made in the intended material. This would make it very sturdy with high accuracy. Nevertheless, that would be very costly and would place high demands on the parts being correct dimensions and design early on, making it harder to make changes as new ideas arise.

Because of the time constraint, the prototype that was created aimed to test the movement of the detector, instead of creating the whole product. The movement of the detector has the same principle of movement as for the light source and same or similar components are used for both movements. If we managed to test the movement for the detector, it is reasonable to conclude that the movement for the light source would work accurately as well. The prototype showed that the movement of the detector could move spherically with the desired angular step, hence the construction of the movement is approved. Although, a few alterations are needed to fulfill the movement of the whole sphere.

The advantages of our concept are the flexibility in the mechanical system through the stepper motor and gears, which makes it easy to change the angular step of the optical components. Since the stepper motor provides such high control of the movement and the flexibility of the system is in place, we know the concept has the ability to measure diffuse light accurately. The concept is fully automated, with 4 stepper motors providing movement in the desired directions. It has a low risk of human error because of the low interaction with the device. It is portable and relatively small in comparison to other products on the market, while still having a steady construction. It has a low cost and has the possibility to be easily implemented in a service. Although, the usability needs to be further considered in the external design.

The disadvantages of our concept are the complexity of the construction. The concept in itself is straightforward and easy, but the implementation of the idea was more complex than expected. It needs to be created properly in its right material and assembled before evaluating if it is realizable. Although, we were aware from the beginning that using a goniospectrometer as a base for development would result in complex constructions. Especially, when a requirement were to conduct spherical movements for the light source and detector. If we make a relative comparison between our construction and other constructions on the market that conduct spherical measurements, other constructions are much more complex and expensive than ours. In that way, the complexity might not be a disadvantage after all.

6.2 Conclusions

The purpose and goals of the project were to develop an optical scattering measurement device, the goals were partially reached because of the reasons explained in section *6.1 Discussion.* A conceptual solution was created that fulfilled the requirements of the project. A prototype was made to validate the movement of the detector which was successful.

Further work is needed to refine the concept and validate its components.

7 FUTURE WORK

In this chapter, recommendations on more detailed solutions and future work in this field are presented. The future work regarding usability, repairability, design of device as well as the service is explained here.

7.1 Usability

From the interviews as well as visualized in sketches, usability and minimizing the possibility of human errors was important. The final product does not have an ideal way of interacting with the separating screen. This is due to a time constraint and should be developed further. The material sample holder needs to be attached to the separating screen with screws and it can be a bit of a hassle since the screws are fairly small and can easily get lost. A better way would be to have a way of clicking it in and taking it out without the use of screws. A challenge with this is to make it flat so that it does not disturb the measurement when light bounces from the material sample.

A touchscreen will be used in order for the user to interact with the device. There is therefore a need to develop the user interface of the screen with consideration to the results from the user study. The interface should include faulty messages, feedback and measurement settings among others.

7.2 Repairability

For sustainable purposes it is important to make the device easy to repair and switch out components when updated. The detector holder and light source holder should be made to be adapted to different kinds of dimensions, they are currently fit to a set diameter. The holders should also be made to safely secure the optical component, so that they do not move from the holder since that has negative consequences on the measurement data. The mounting of the detector and light source should have an interface to enable the exchange and update of these components.

7.3 Design of the device

The material sample holder uses nuts and screws as fasteners. This is due to the choice of manufacturing. It is hard to 3D print a thread and the nut acts as one instead. If the material sample holder is manufactured in another way, the hole can be threaded instead to minimize the amount of components. The total thickness of the material sample together with the sample holder needs to be 20 mm to fit seemingless with the separating screen. To accommodate for different thickness of material samples, there could be additional add-ins with different thickness that could be added between the material and holder to achieve the desired thickness.

Currently, the four motors in the device are not covered from the rest of the components. Since the motors are being purchased from a retailer and are not manufactured, they do not meet the demands of not reflecting light. Therefore there is a need to cover the motors in an outer light absorbent shell, preferably that is easy to take off to enable easy repairs if needed.

The motors that drive the detector and light source in the azimuthal direction are placed vertically, this was due to the time limit. By instead placing the motors horizontally and using a bevel gear, the axle and bearing could have a slightly lower height and would in turn affect the total size of the device. If placed horizontally there will be no need of making a cutout for the

encoder and the platform where the motor stands on will therefore be more durable. Further, it is important to develop a secure way of fastening the motors on the platforms.

One issue that has not been prioritized is the cords from the detector and light source as well as the cords from the motor associated components. It is important that the cords have an assigned path to move on so that they do not get stuck in each other and disturb the measurement.

The electrical components besides the motors such as encoder, motor driving unit and power supply will be placed at the bottom. The components are fairly big and are therefore placed at the bottom instead of beside the motor that drives the φ-direction. The bottom part of the device will therefore need to be developed so that the electrical components are organized and have a designated position but at the same time can easily get repaired. The angle of incident light is controlled by a motor that is attached to a gear. If the angle was controlled manually this would reduce the need of motors, connecting gears and other components. This would lead to a reduced cost of the device. Since the light source does not move during measurements it is an area that is reasonable to investigate further.

The detector can not move freely 180°. This has to do with the size and placement of the detector. If the detector holder was placed inside of the internal gear it would move without problem, however that would lead to the separating screen needing to be significantly smaller and that would lose the purpose of having the separating screen, which is to limit light contamination. Another way of allowing the full movement would be to make the detector and detector holder smaller as well as making the shift of the internal gear even bigger from the axle, allowing it to pass through. Further, a solution to allow for the full movement of 180° would be to make the supporting construction and internal gear bigger, but that would result in a device that is significantly larger.

Solid mechanics calculations need to be performed on the supporting circular construction, the motor platforms and axle parts. This to ensure that the components are durable as well as not overdimensioned. The calculations regard deflection and material fatigue. Moreover, the bearings need to be calculated based on the force and angular velocity in order to choose appropriate materials and dimensions. It is recommended to perform tests on whether or not the bearings need lubricant or not for smoother movement.

As mentioned in *4.1.3 Movement of detector,* the fixating structure of both the detector and light source has a shift to the side compared to the axis through the axle parts. The light source and the detector have the same diameters, but since the internal gear and fixating structure has a bigger thickness to withstand the big dimensions and since there is no limit the thickness such as the detector movement has, there is a need for a bigger shift regarding the light source. The bearing that connects to the fixating structure therefore has to be prolonged at the connecting part. This so that the light source will be placed as if it was in the middle plane pointing at the material sample.

There is a need to investigate the thickness of the detector track to minimize blindspots. The current detector track is dimensioned in regards to the light source movement with an angular increment of 5°. The detector track is at its maximum thickness for the avoidance of blindspots, but it is also interesting to see what the minimum thickness is in terms of material usage. The thinner the construction, the less use of material. The calculations that need to be conducted are solid mechanics; deflection of the material platform, gear and the supporting construction.

Before manufacturing the device in the intended material discussed in *4.2 Material selection*, tests need to be performed to see if the matte surface achieved through etching meets the

reflectivity requirement. Different levels of etching can be tested for the reflectance to find the optimal one.

The axle parts that connect at the bottom and at the top need to be designed so that they can be secured into a stable part. The axle cannot be made in one whole piece, instead, it has to be made in singular parts since the bearings need to be mounted one at a time.

The top and bottom of the axle needs to be secured into a stable part or a frame. There need to be separate parts to enable correct assembling and also due to manufacturing aspects. But when assembling them, they need to be fastened to the floor and roof of the device. The external design therefore needs to include a frame on the inside where the top and bottom axle parts can be fastened.

7.4 Service

Since the device will be provided as a service there is a need to provide an adequate online booking platform for the intended customers. A platform should therefore be developed and could be added to AFRYs webpage for easy access. The platform should include an easy way of booking and also an explanation of what the service offers and potential areas of use. By doing this it is easy to attract new customers that visit AFRYs webpage and get intrigued by the service.

The user interface of the device needs to be made with usability in mind to enable different levels of users.

If a part needs reparation or a part needs to be exchanged, a system of collecting these needs to be in place before offering the service to customers.

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APPENDIX A. WORK BREAKDOWN STRUCTURE

Figure 1 shows the finalized work breakdown structure (WBS).

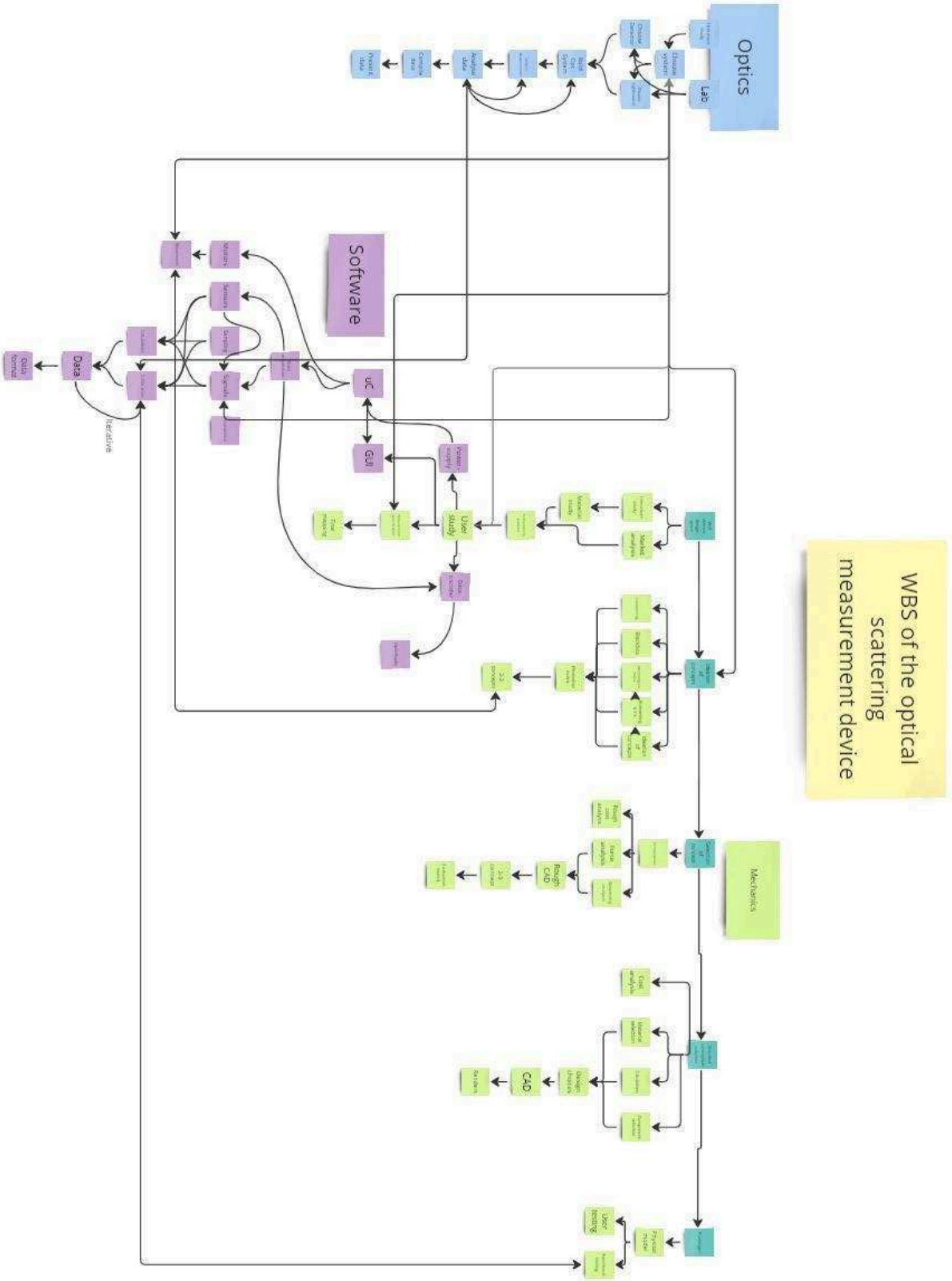


Figure 1. WBS

APPENDIX B. GANTT-SCHEDULE

Figure 1 shows the GANTT-schedule as a whole.

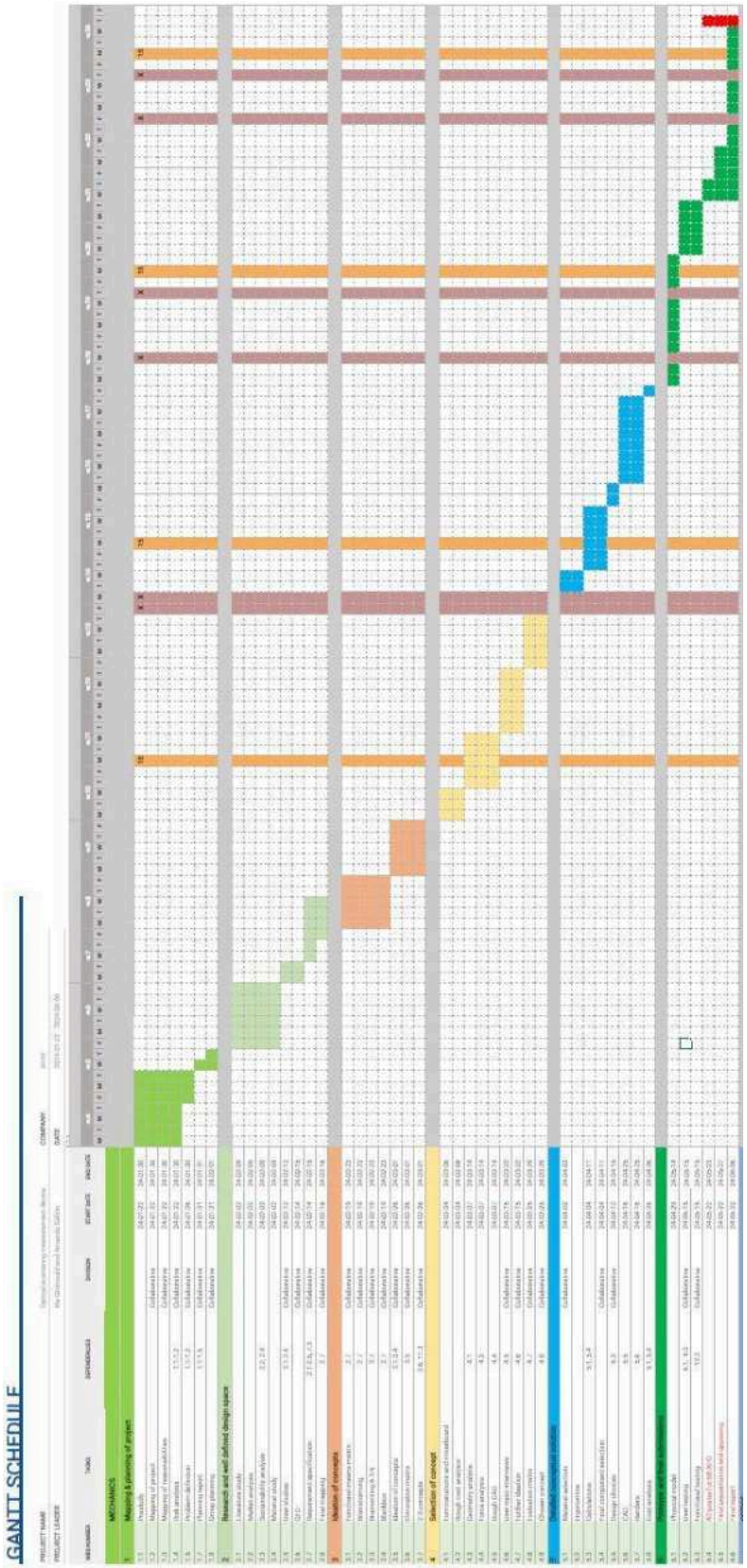


Figure 1. GANTT-schedule

APPENDIX C. INTERVIEW GUIDE

The interview guide is presented below.

Background

What is your education and what do you work with today? How much knowledge do you have about optics?

Previous experience

Have you used similar devices before?

If no:

How do you simulate materials as of today?

What parameters are you looking for then?

How reliable is the result?

How many measurement points are required to give a reliable result, in your opinion?

If yes:

What kinds of machines/ brands have they been?

Were there clear instructions on how to use it? Training?

Do you feel safe using the machine? What is contributing to it?

If feeling insecure, what is contributing to it?

What did you like about the machine?

What was there that you didn't like about the machine?

Was there feedback confirming proper use or if something went wrong?
If not, would you like to have feedback?

What is the problem with the products on the market today?

How long time did it take to make a measurement?

When will you use the machine? For what area of use?

What is easy? What is difficult?

How many measurements are made on the same material sample? Why?

Information regarding the product

Calibration, how is that done today?

As we understand it, it is important to take into account the absorption rate of the material, the color and the hygiene of the surrounding environment, as it can interfere with the measurement. Why is it like that?

Future device

How often will the product be used? On what occasions?

For what area of use?

What kind of interface do you prefer?

How would you like to receive information from the machine, e.g. error messages/proper use?

How much should the user be able to do? For example locks in case of misuse.

Is it important to be able to keep settings for the next measurement, to have some kind of preset that does not need to be changed at each measurement?

How often does the position of the components in the measuring tool change? For example, the material sample holder may need to be changed often when changing the material sample, while the light source may only be changed once.

If it breaks? Who will repair it?

If something goes wrong, how interactive should the machine be in how to correct the error?

Would you consider repairing it if instructions and tools were available?

Are there standards to adhere to? (Measuring instrument standards in optics and handling of lasers)

Which parameters do you want to enter? What do you want to have control over?

What do you think about the noise level of the machine? Is there anything that would disturb?

What settings would you like to make in a possible menu before the measurement?

How important is it that it is possible to quickly learn a new measuring instrument?

Dream scenario? What would it look like in that case.

Questions that were added along the way

Tell us what you know about gonioreflectometers. Do you have a lot of knowledge about them?

Strengths vs weaknesses with gonioreflectometer?

Have you used a gonioreflectometer before?

What should we take into account regarding the surrounding environment of the optic measurement?

What wavelengths do you usually measure in? How often do you measure samples in UV light?

How important do you think it is to be able to replace the components? And how often will it be done?

APPENDIX D. REQUIREMENT SPECIFICATION

Figure 1 shows the requirement specification of the project.

Requirement	Class	Evaluation
1. Main requirement		
1.1 Measurement of optical scattering	MR	Specular and diffuse light
2. Basic requirements		
2.1 Measure reflection and transmission	R	
2.2 Measure NIR, VIS, UV	R	Wavelength between 270, 380-700, 950 nm
2.3 Entail replacement of components	R	Lightsource and detector
2.4 Allow measurement of anisotropic materials	R	
2.5 Enable spherical measurement	R	4π
2.6 Entail measurement of polarized light	R	Polarized light is the geometrical orientation of the lightwave oscillations
2.7 Allow for distance between sampleholder and detector	R	Maximum 200 mm
2.8 Allow for distance between sampleholder and lightsource	R	More than 200 mm
2.9 Avoid intersection between detector and lightsource when measuring	R	
2.10 Provide no movement in material holder	R	No rotation or translation during measurement
2.11 Provide no movement of lightsource during measurement	R	
2.12 Allow movement of lightsource between measurements	R	0-85 degrees in θ-direction in spherical coordinates
2.13 Allow for adabtable angular increment of lightsource	R	Minimum 5 degrees
2.14 Allow for adaptable angular increment of detector	R	Minimum 2 degrees in both θ- and φ-direction
2.15 Allow for color of internal components that provide no reflection	R	Within wavelength measurement range
2.16 Provide stability in static and dynamic components	R	
2.17 Fit different material samples	R	Maximum dimension of 50 x 50 x 10 mm
2.18 Allow for weight of detector	R	Maximum 300 g
2.19 Allow for weight of lightsource	R	Maximum 300 g
2.20 Allow for outer dimensions of lightsource system	R	Maximum 120 x 50 mm
2.21 Minimal human interaction before and after measurement	R	
2.22 No human interaction during measurement	R	
2.23 Enable easy cleaning	R	Of internal environment, sample holder, lightsource and detector
2.24 Maintain desired humidity for optical components	R	
2.25 Maintain desired temperature for optical components	R	10-30 degrees
2.26 Be time efficent	W	
2.27 Maximize hygiene	W	

3. Usability		
3.1 Feedback for correct/incorrect use	R	Considered by project members responsible for the software
3.2 Easy to use	R	
3.3 Minimize human error	R	
3.4 Minimize risk of misuse	R	
3.5 Allow for repeatability of measurements	W	
3.6 Provide preferred actions in case of error	W	Considered by project members responsible for the software
3.7 Allow for customized settings	W	Considered by project members responsible for the software
3.8 Allow for different kinds of users	W	

4. Ergonomics and safety		
4.1 Follow optical component directives	R	
4.2 Clear indications when in use	R	
4.3 Provide easy accessability of sample holder between measurements	R	
4.4 Provide accessebility of lightsource and detector	R	
4.5 Fit different handsizes	W	5-95th percentile of population

5. Sustainability		
5.1 Be repairable	R	Sample holder, detector holder, light source holder, environment and openi
5.2 Design for circular readiness	W	
5.3 Limit critical materials	W	Use only when no other options are available
5.4 Be recyclable	W	
5.5 Allow for high material hygiene	W	In components and preferably between components
5.6 Optimize design for reuse of components	W	

6. Production and mounting		
6.1 Enable customization of optical components	R	
6.2 Enable modularity	R	
6.3 Enable repairs of mechanical componenets	W	
6.4 Enable exchange of optical components	W	
6.5 Minimize production cost	W	
6.6 Design for assembly	W	

7. Market and target group		
7.1 Express AFRV values	R	Brave, devoted and team player
7.2 Express expertise	R	Within the field of optics
7.3 Optimize design for market competition	W	Usability, brand values and area of application

Figure 1. Requirement specification

APPENDIX E. QUALITY FUNCTION DEPLOYMENT

Figure 1 shows the Quality Function Deployment of the project.

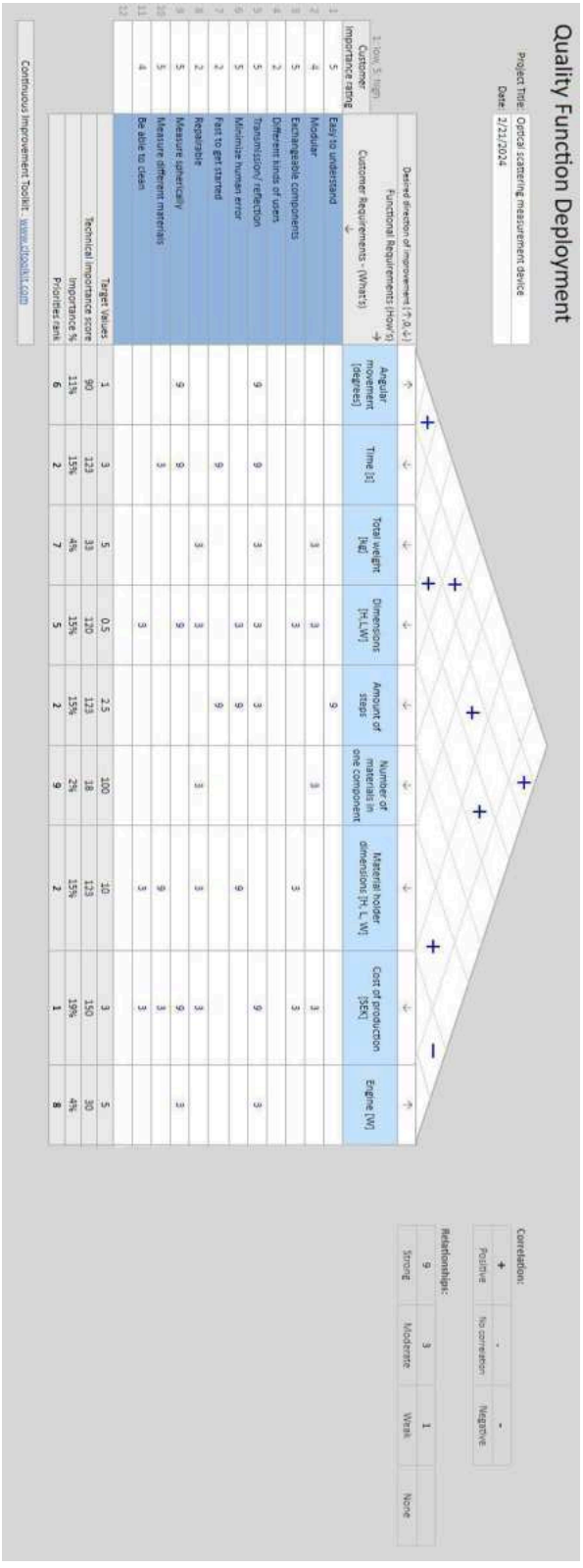


Figure 1. Quality Function Deployment

APPENDIX F. SKETCHES

Figure 1 shows the sketches of the spherical movement. Figure 2 shows ideas for the material sample. Figure 3 shows ideas of how to access the material sample. Figure 4 shows the early design ideas of concept 1. Figure 5 shows the early design ideas of concept 2.

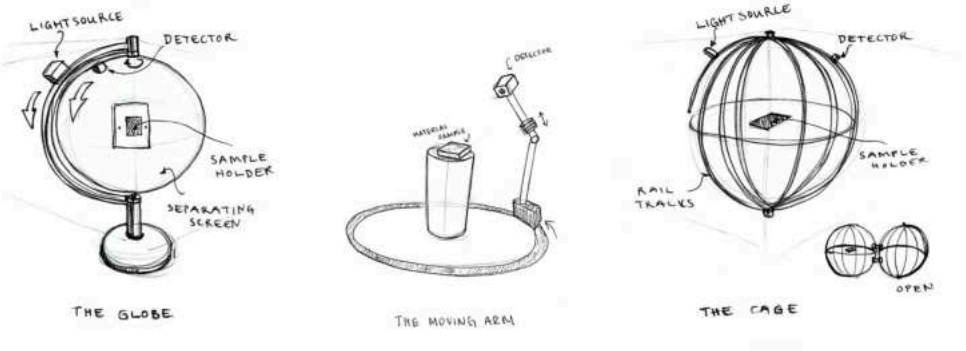


Figure 1. Spherical movements

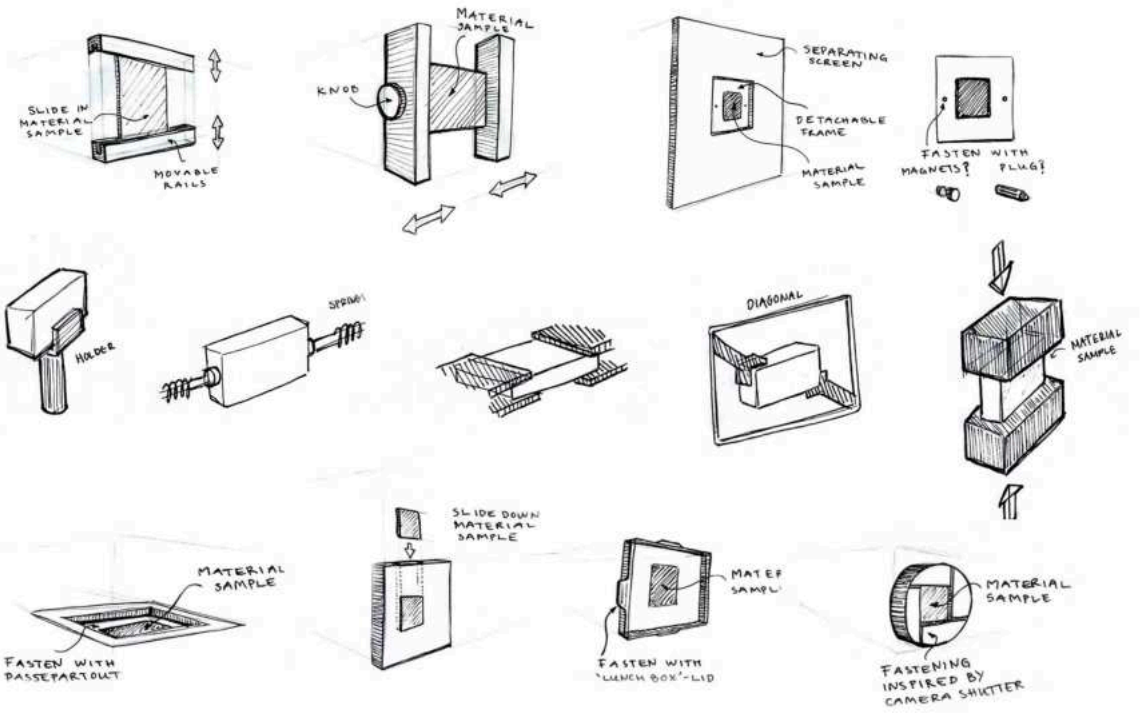


Figure 2. Sketches of ideas for material sample holder

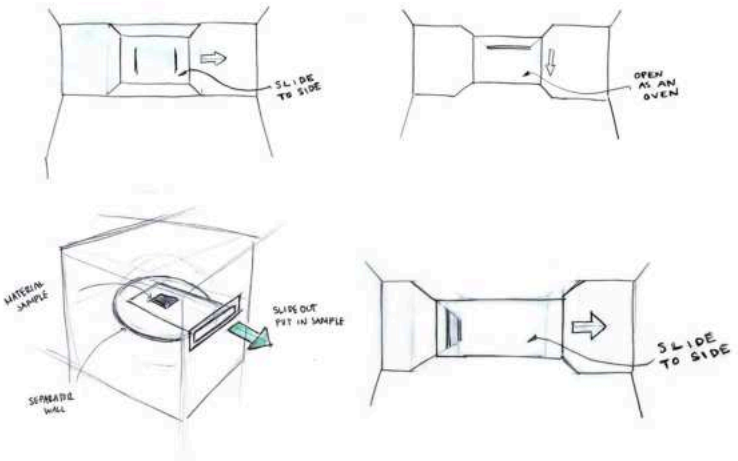


Figure 3. Ideas of how to access material sample holder

APPENDIX G. MORPHOLOGICAL MATRIX

Figure 1 shows the morphological matrix of the project.

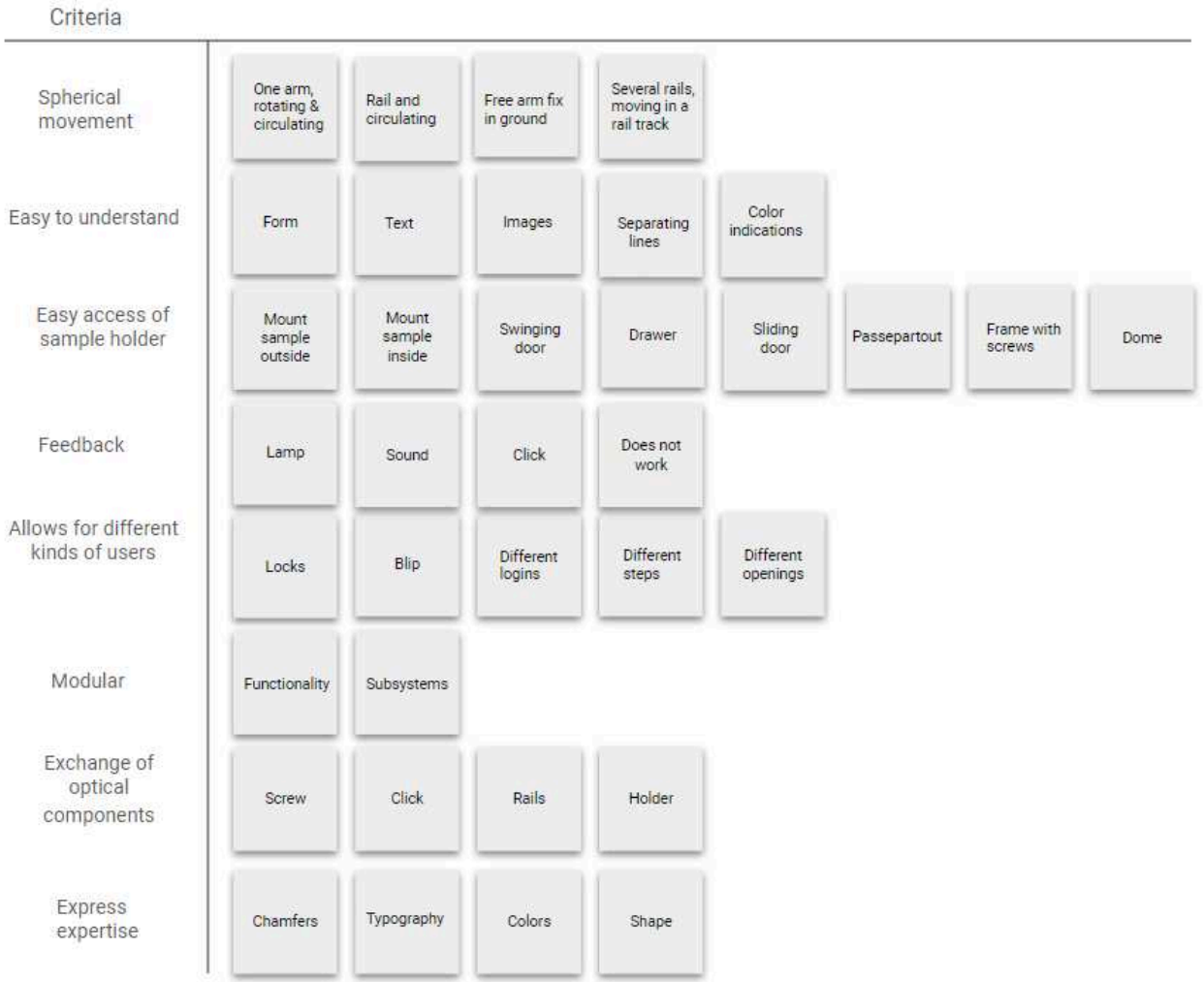


Figure 1. Morphological matrix

APPENDIX H. VARIATIONS

Figure 1 shows form variations of concept 1 and Figure 2 shows variations of concept 2.

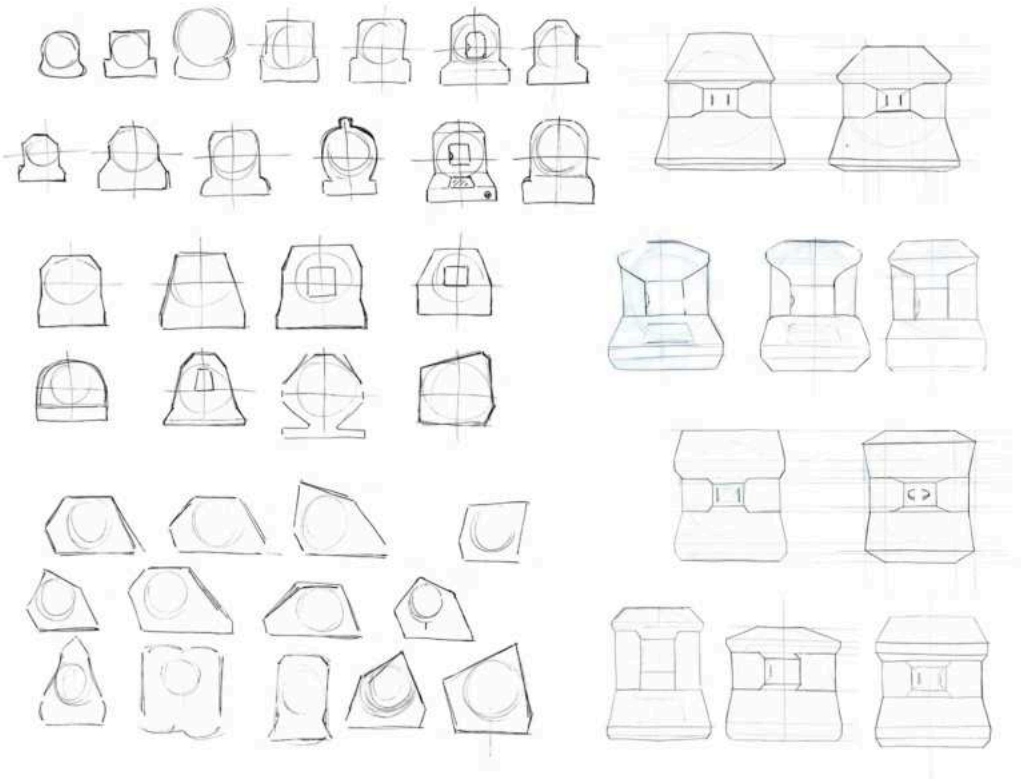


Figure 1. Iterations of Concept 1

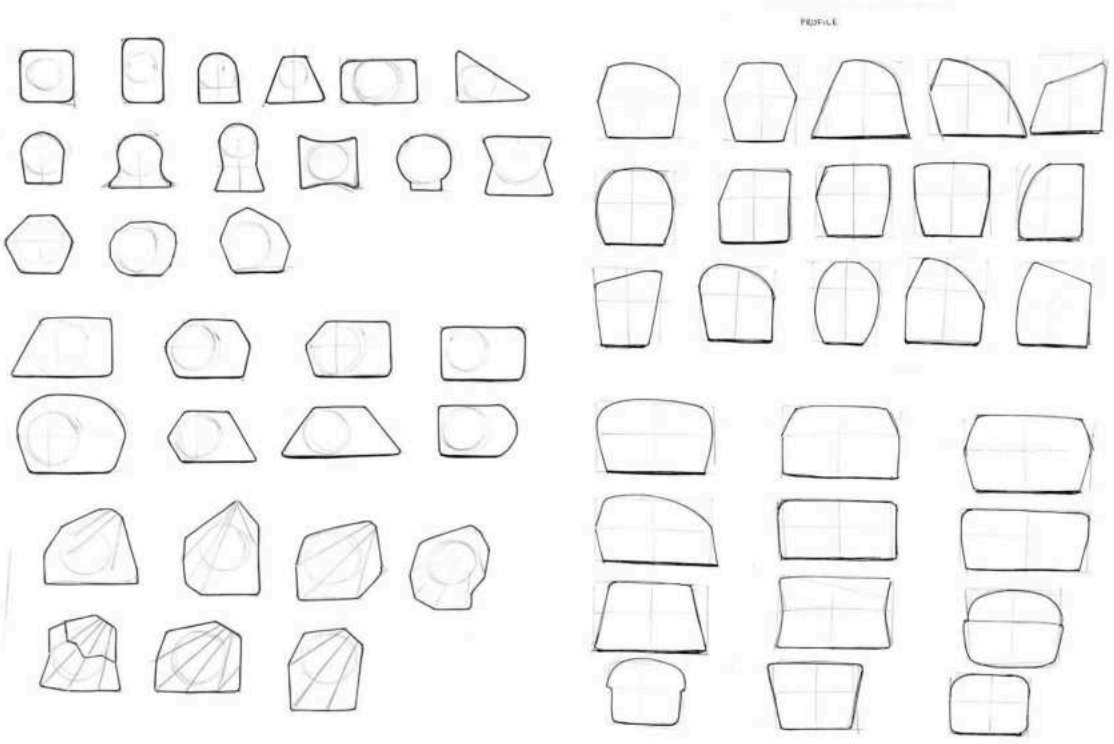


Figure 2. Iterations of Concept 2

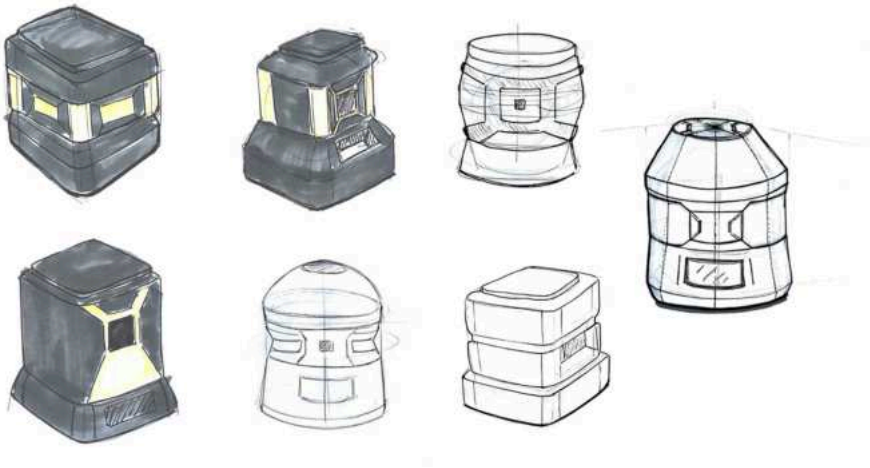


Figure 3. Early design ideas Concept 1

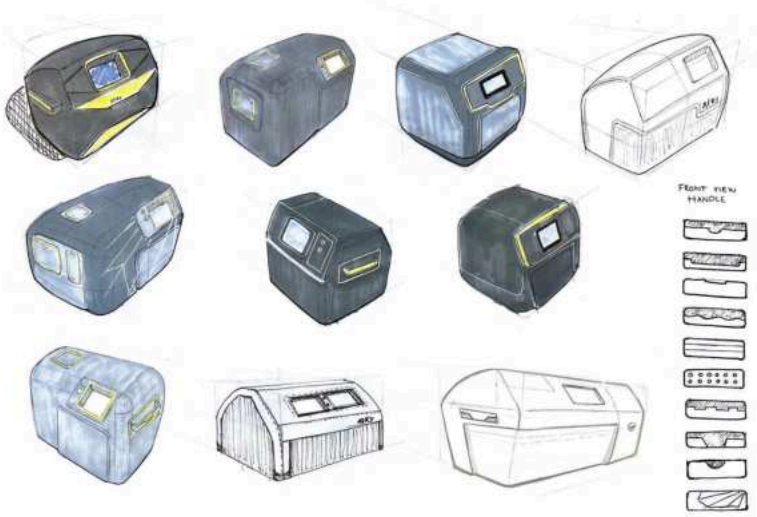


Figure 4. Early design ideas Concept 2

APPENDIX I. BILL OF MATERIALS

Table 1 shows all the CAD components of the optical scattering measurement device as a bill of materials.

Table 1. Bill of Materials

ITEM NO.	PART NUMBER	QTY.
1	SEPARATING SCREEN	1
2	AXLE_TOP_DETECTOR	1
3	AXLE_BOTTOM_DETECTOR	2
4	FIXATINGSTRUCTURE_OUT_FASTENINGBEARING	1
5	AXLE_BOTTOM_LIGHT	1
6	FIXATINGSTRUCTURE_IN_MOTORPLATFORM	1
7	DETECTORINTERNAL GEARS	1
8	AXLE TOP LIGHT	1
9	MATERIAL SAMPLE HOLDER TOP	1
10	MATERIAL SAMPLE	1
11	DETECTOR	1
12	MOTORGEARS	1
13	INTERNAL GEAR LIGHTSOURCE	1
14	BEARING_GEAR_OFFSET	2
15	FIXATINGSTRUCTURE_FASTENINGBEARING_LIGHTSOURCE	1
16	FL57STH56-2804UEN-OEM-MOTOR	4
17	DOUBLEGEARS_MOTOR	2
18	DETECTOR_HOLDER	1
19	GEAR_MOTOR	2
20	FIXATINGSTRUCTURE_MOTORPLATFORM_LIGHTSOURCE	1
21	MOTOR HEIGHT	1
22	BEARING TOP OFFSET	2
23	LIGHTSOURCE_HOLDER	1
24	LIGHTSOURCE	1
25	MOTOR_GEAR_PINION_LIGHTSOURCE	1
26	M5 SCREW	19
27	SCREW MATERIALSAMPLE	10

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